University College Dublin (UCD)
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An Evaluation of ESA Biomass Mission Data for Ocean and Ice Applications.

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Name: Orlaith Doyle (Ní Dhúill)
Supervisor: Björn Rommen
Student Number: 19424364
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Statement of Authorship

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Declaration and description of AI use

Prompts & tools used to assist in completion of this work can be found in Appendix-C.

Signature:

Acknowledgments

I wish to thank all who have helped me so far in my education and career, it definitely takes a team! Firstly, I wish to express deep gratitude to my internship supervisor, Björn Rommen, for giving me the extraordinary opportunity to experience life at ESA. I am very grateful not only for his valuable mentorship, but also for allowing me to live-out a lifelong dream of working with ESA. I also thank the entire team within ESTEC's Earth Observation division, my office peers, and the wider organisation as a whole, for their kindness from day one. I am grateful too to Kostas Papathanassiou and the team at DLR for sharing their expertise and invaluable insights through weekly classes in Biomass data handling, which provided essential foundations and enabled me to build-on and adapt my remote sensing experience to the Biomass mission.

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Abstract

Biomass, ESA's first P-band SAR satellite, was launched in April 2025 to measure global forest Biomass. This project tests a secondary application: using P-band data for ocean remote sensing. The main question is whether HH–VV copolar phase difference (CPD) can be used as a proxy for sea surface salinity (SSS). The physical basis is that salinity affects the dielectric constant manipulation chain established in §2. A secondary, less developed strand looks at whether Biomass acquisitions can be used for wave and sea-state characterisation; this is still in progress and reported here only briefly.

A multi-stage pipeline was built to extract, calibrate, and geolocate Biomass SLC products, mask land regions, and compute circular phase statistics over open ocean. Results were checked against GLORYS12 reanalysis and SMAP L3 salinity across four regions, including the Ganges-Meghna and Amazon plumes.

The results of this research found the data roughly supports CPD as an early SSS retrieval method under the conditions tested based on related correlation. Ionospheric phase noise, sea-state roughness, and wind are identified here as the main confounders still to be constrained. Biomass was not designed for ocean science, so this work is a first check on whether it has any use there, plus a pipeline potential future interns or researchers can use in their work.

Future work expected to be completed during the remainder of this internship includes research in cryospheric applications including groundline detection and potential subglacial lake studies beneath the Antarctic shelf.

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Acronyms

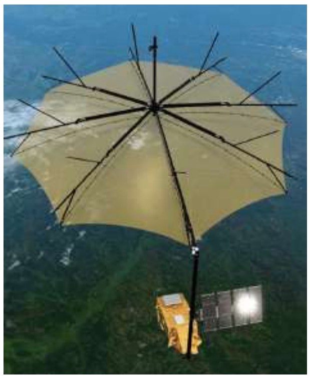
CDF	Concurrent Design Facility
CMEMS	Copernicus Marine Environment Monitoring Service
CPD	Copolar Phase Difference
DEM	Digital Elevation Model
DOA	Date of Acquisition
DOI	Date of Interest
DLR	German Aerospace Center (Deutsches Zentrum für Luft-und Raumfahrt)
ECMWF	European Centre for Medium-Range Weather Forecasts
EE	Earth Explorer
EO	Earth Observation
ESA	European Space Agency
ESTEC	European Space Research and Technology Centre
GIS	Geographic Information System
GLORYS	Global Ocean Reanalysis and Simulation
IEM	Integral Equation Model
InSAR	Interferometric Synthetic Aperture Radar
MAAP	Multi-Mission Algorithm and Analysis Platform
NRCS	Normalized Radar Cross Section
OSTIA	Operational Sea Surface Temperature and Ice Analysis
P-band	P-band Microwave Frequency Band
PolInSAR	Polarimetric Synthetic Aperture Radar Interferometry
PolSAR	Polarimetric Synthetic Aperture Radar
PSU	Practical Salinity Unit
QGIS	Quantum Geographic Information System
ROI	Region of Interest
SAR	Synthetic Aperture Radar
SLC	Single-Look Complex
SMAP	Soil Moisture Active Passive
SMOS	Soil Moisture and Ocean Salinity
SNAP	Sentinel Application Platform
SNR	Signal-to-Noise Ratio
SSS	Sea Surface Salinity
SST	Sea Surface Temperature
STAC	SpatioTemporal Asset Catalog
UCD	University College Dublin
WP	Work Package
YPSat	Young Professional Satellite

1 Introduction

This report documents research completed during a six month internship within the Earth observation directorate at the ESTEC in Noordwijk, the Netherlands. As the technical centre of ESA, ESTEC is the hub where satellite missions are introduced, designed, and managed. Working within this environment provided direct access to expert teams, hands-on experience, continuous guidance and use of data systems that support Europe's leading teams in space science, and in particular, Earth observational climate sciences and remote sensing.

This internship project combined scientific research, Earth observation data analysis, and computational model development to investigate novel scientific applications of data acquired by the ESA's P-band Biomass mission[1]. Building on theoretical knowledge gained through UCD coursework and supported by ESTEC's multidisciplinary research environment, the project explored how the mission's unique P-band SAR observations could be used beyond their primary forestry objectives. Particular emphasis was placed on evaluating the potential of Biomass for applications in oceanographic and cryospheric remote sensing, which would contribute to the broader scientific exploitation of ESA's newest Earth Explorer mission.

(a) ESA Earth Explorer missions in order of launch, with Biomass as the seventh mission.



(b) Biomass Payload from above.

Figure 1: ESA EE family [2] (left) with artist visual of payload in flight (right)[1].

1.1 State-Of-The-Art: Biomass, The First P-Band EO Mission

Following its launch aboard a Vega C rocket from Kourou in April 2025, ESA's Biomass EE satellite began collecting raw radar data to help our understanding of the global carbon cycle [3]. Manufactured by Airbus Space and Defense, it features a large 12-meter deployable reflector antenna powering the first-ever space P-band Synthetic Aperture Radar (SAR). Biomass is special as it operates at a much longer wavelength than standard satellite radar tools (at approximately 70cm), the Biomass signal achieves unprecedented and unrivaled penetration depth. While the mission's primary operational objective is to measure global forest height and forest Biomass over its 5-year lifetime, its long P-band wavelength also opens opportunities to investigate applications beyond forest Biomass, including oceanographic and cryospheric processes. Over ice, the increased penetration depth offers sensitivity to subsurface structures, while over the ocean the longer wavelength provides a different interaction with the sea surface than conventional C-band SAR, creating opportunities to investigate dielectric and wave-related scattering processes. [4, 5].

1.2 Biomass Mission Secondary Objectives: Cryosphere and Oceanic

Traditional space-based satellites and Earth observation work done by other competitors (NASA.EO, GEO, ISRO.EO, IceEye, Satellogic and additional monitoring tools) have left Earth observational scientists essentially *blind* to subsurface earth dynamics. Standard optical cameras and high-frequency microwave radar sensors (such as X-band or C-band) scatter immediately upon hitting surface interfaces. They capture only what can be observed on top of the planet's surface [6]. During this internship, this limitation was evaluated across two very different and dynamically challenging surface domains:

- **Oceanography:** Shorter radar wavelengths are generally blinded by ocean foam, wave spray, heavy atmospheric interference during extreme weather events and in some cases; constant wave motion. [7]. This makes it very challenging to detect deep subsurface dynamics like marine internal waves or ocean salinity boundaries where fresh meltwater plumes hit dense seawater [8, 9].

- **Glaciology:** Standard sensors are heavily disrupted by surface snowfall and meltwater. They physically cannot look deep into the ice sheet to map internal structural accumulation or the underlying bedrock topography. These are the key factors that influence ice velocity and ice sheet stability.

These additional applications are interesting to both the Earth observation and wider space science communities, as they explore whether a satellite P-band SAR system designed primarily for forest Biomass estimation can also be repurposed to investigate cryospheric and oceanographic processes. Owing to its long wavelength and fully polarimetric capability, Biomass provides a unique perspective compared with the C-band SAR systems commonly used for operational marine remote sensing, offering sensitivity to different scattering mechanisms and environmental processes. Expanding the scientific applications of Biomass beyond its primary mission objectives is therefore important for maximising the mission's overall scientific return while helping provide new information for development of future Earth observation missions.

1.3 Internship Objectives and Scope

This internship thesis focuses on the cross-disciplinary utility of Biomass P-band SAR data. The project addresses three primary objectives:

1. **Investigate Oceanographic Applications of Biomass P-band SAR**: Evaluate the sensitivity of Biomass P-band polarimetric observables to sea surface salinity and temperature, while assessing how ocean roughness, swell, and radar viewing geometry influence these measurements. A secondary objective was to assess if the same observations could characterise ocean wave and sea-state properties.[10, 11].

2. **Evaluate Cryospheric Observation Capabilities of Biomass P-band SAR**: Examine the ability of Biomass P-band signals to penetrate snow and ice and investigate their potential for detecting and characterising subsurface cryospheric features, including grounding line locations, internal ice structures, subglacial hydrological systems, and potential subglacial lakes.

3. **Assess Broader Earth System Science Opportunities**: Evaluate opportunities to extend Biomass applications beyond Biomass retrieval by integrating P-band observations into oceanographic and cryospheric studies of climate-driven processes.[12].

By investigating these topics under the mentorship and guidance of mission expert and Biomass Mission Scientist, Björn Rommen, this research highlights how the Biomass mission can extend its reach far beyond its forest goals and delve even *deeper* into the Earth's hidden ocean and ice subsurface systems.

2 Scientific Theory

Biomass is a fully polarimetric, single-frequency P-band (435 MHz, $\lambda \approx 69\,\mathrm{cm}$) SAR and first of its kind to operate at this frequency [13]. Four questions structure the background theory that follows:

1. How Biomass measures electromagnetic scattering (§2.1),
2. Why that measurement should be sensitive to SSS (§2.3) and how ocean waves may complicate that sensitivity,
3. If the same SAR image can be used to characterise them directly (§2.2-§2.4),
4. Why the P-band frequency is of particular interest beyond the ocean, motivating the future-work directions in §2.5.

2.1 Polarimetric SAR Fundamentals

NOTE: This subsection touches on only the relevant core principles and equations used within this project. Greater detail can be found within the code and commentary in Appendix Section B. For a broader introduction to SAR, see ESA's dedicated courses [14].

A fully polarimetric SAR system can transmit and receive signal in both horizontal (H) and vertical (V) linear polarisations, recording the complex scattering matrix for every resolution cell, where each element is a complex number encoding the amplitude and phase of the backscattered field for a given transmit-receive polarisation pair.

$$\mathbf{S} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \tag{1}$$

On a relatively *flat* reciprocal medium such as the ocean surface, $S_{HV} = S_{VH}$. This allows three independent complex observables; S_{HH}, S_{VV}, S_{HVorVH}. It's generally more convenient to work with the Pauli scattering vector from which the 3×3 coherency matrix; $T = \langle kk^{*T} \rangle$ is created by averaging or multilooking across neighbouring pixels.

$$\mathbf{k} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{HV} \end{bmatrix} \tag{2}$$

The Cloude-Pottier eigen-decomposition of T (where entropy is denoted by H, anisotropy by A and mean alpha angle of α) provides a target-decomposition framework used elsewhere in this project for scene classification [15]; the ocean work in this chapter instead uses two narrower observables drawn directly from the covariance matrix. The first is the interferometric-style coherence between the HH and VV channels, which is used as a quality gate; high HH-VV coherence over open ocean indicates the two channels are seeing the same physical scatterers, so their relative phase is meaningful as opposed to completely de-correlated noise.

$$\gamma_{HH,VV} = \frac{|\langle S_{HH} S_{VV}^* \rangle|}{\sqrt{\langle |S_{HH}|^2 \rangle \langle |S_{VV}|^2 \rangle}} \tag{3}$$

The second is the co-polarisation phase difference (CPD), which is the primary salinity observable developed in §2.3, alongside the amplitude ratio $\beta = |S_{HH}|/|S_{VV}|$, used interchangeably with φ in this work as a second, amplitude-domain observable sensitive to the same Fresnel asymmetry.

$$\varphi = \arg(\langle S_{HH} S_{VV}^* \rangle) = \mathrm{wrap}(\varphi_{HH} - \varphi_{VV}) \tag{4}$$

Both are computed after multilooking or spatial averaging of T, to minimise speckle, the multiplicative noise is intrinsic to coherent imaging, at the cost of spatial resolution.

2.2 Why P-band?

Biomass has three properties that make it distinctive for the applications considered within this research and future applications discussed below, none of which is possible using Sentinel-1 (C-band, 5.6 cm) or any other operational SAR data.

- **Penetration abilities;** Attenuation through a dissipative medium like vegetation, dry snow or sand scales strongly with frequency, so P-band penetrates far deeper than C-band or even L-band before the signal is fully attenuated. This is the physical basis of the primarily forest focused mission, and also the basis for the subsurface applications in §2.5.
- **Bragg-scale mismatch;** As shown below in *Eq. 5*, the resonant ocean roughness order scale at P-band is in metres instead of centimetre-scale ripples as in C-band and L-band ocean SAR imaging. This means P-band ocean scattering is sensitive to a different, longer part of the ocean wave spectrum than older SAR wave-mode products, and that the extensive C-band modulation-transfer-function literature used operationally by Sentinel-1 Wave Mode does not transfer directly, which is a limitation discussed further in §2.4.
- **Dielectric Sensitivity;** As mentioned in §2.3, the low-frequency conductivity term in the Debye relaxation model *(Eq. 6)* makes the P-band dielectric constant of seawater far more sensitive to salinity than at L-band, in principle improving on the sensitivity already exploited operationally by SMOS and SMAP [13].

Together, these three properties are why a mission designed primarily to weigh the world's forests may also, as an experimental by-product, motivate new ocean and subsurface science.

2.3 Ocean Scattering Physics

Two physical chains connect the raw polarimetric observable data of §2.1 to geophysical quantities of interest, and they share a common link: the sea surface. The first chain explains why the surface scatters at all; the second explains why the scattered signal carries a salinity signature. Ocean backscatter at radar incidence angles away from nadir is dominated by *Bragg scattering* [7]: resonant returns from short surface-roughness components whose wavelength (Λ) satisfies the Bragg condition;

$$k_B = 2k_r \sin\theta, \qquad \Lambda = \frac{2\pi}{k_B} = \frac{\lambda_r}{2\sin\theta} \tag{5}$$

Here, k_r and λ_r are the radar wavenumber and wavelength respectively, and θ is the local incidence angle. At P-band, the roughness scale that dominates Biomass ocean returns is physically different from that seen by conventional ocean-imaging SAR systems.

The amplitude of the Bragg-resonant return is itself modulated by longer gravity waves through three coupled mechanisms which are the physical basis of the wave-imaging observables developed in §2.4.

Tilt Modulation; The local incidence angle varies over the long-wave slope, changing the effective Fresnel reflectivity,

Hydrodynamic Modulation; Short Bragg waves are compressively bunched near long-wave crests by the orbital straining of the surface current.

Velocity Bunching; The orbital velocity of the long wave Doppler-shifts scatterers, displacing them in the SAR azimuth image.

The second chain links salinity to electrical physics, as opposed to geometric, properties of the surface. Seawater is an imperfect dielectric whose complex relative permittivity is described, over the microwave range relevant here, by a single-relaxation Debye model with an added ionic-conductivity term [13]:

$$\varepsilon(f) = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + i2\pi f\tau} - i\frac{\sigma}{2\pi f\varepsilon_0} \tag{6}$$

Where ε_s and ε_∞ are the static and high-frequency limiting permittivities, τ is the Debye relaxation time, σ is the ionic conductivity, and ε_0 is the permittivity of free space. Both τ and σ are functions of salinity S and temperature T, and at P-band frequencies of 0.435 GHz, deep in the low-frequency region of the relaxation spectrum, the conductivity term dominates the salinity sensitivity of ε far more than at higher microwave frequencies, which is the core physical reason P-band was proposed for salinity sensing at all [13]. Recent laboratory measurements have extended the Debye fit specifically to P-band and to salinities up to 138 pss [13], which is the dielectric model adopted in this project's retrieval pipeline in place of the L-band derived coefficients used by SMOS and SMAP.

The dielectric constant enters the observable radar signal through the Fresnel reflection (R) coefficients at the air-to-sea interface, which for the two linear polarisations are as follows;

$$R_{HH} = \frac{\cos\theta - \sqrt{\varepsilon - \sin^2\theta}}{\cos\theta + \sqrt{\varepsilon - \sin^2\theta}} \tag{7}$$

$$R_{VV} = \frac{\varepsilon\cos\theta - \sqrt{\varepsilon - \sin^2\theta}}{\varepsilon\cos\theta + \sqrt{\varepsilon - \sin^2\theta}} \tag{8}$$

Because R_{HH} and R_{VV} respond differently to changes in ε, both their ratio (β, the amplitude analogue of *Eq. 4)* and the relative phase between them (the CPD, φ) carry a salinity signature that is, to first order, independent of the absolute (uncalibrated) radiometric gain of the instrument, a self-calibrating property that motivates using a polarimetric ratio observable as opposed to absolute backscatter σ^0 for salinity retrieval.

2.4 Ocean Applications: Salinity and Waves

This section brings together §2.1-§2.2 into the two observable-generation pipelines actually implemented in this project, corresponding to the oceanic objectives of this internship.

Salinity. The parametric ratio observables of *Eq. 4* are computed per multilooked pixel from Biomass Level-1 SCS products, masked to open ocean, and inverted against a Fresnel/Debye lookup table *(Eqs. 6-8 from [10], [16])* to recover an apparent permittivity, then mapped to salinity via the same Debye fit run in reverse. Because *Eqs. 7-8* assume a flat interface, the inversion is only valid once the surface-roughness contribution to β and φ has been removed via an empirical wind/roughness correction using ERA5 wind data. Retrievals are validated against SMOS and GLORYS12 reanalysis. Salinity ROI are chosen for their strong sea surface salinity-to-freshwater contrast, testing whether the physical relationships of *Eqs. 6-8* are achievable using real Biomass data.

Waves. The same SLC products support a second, independent pipeline built around the tilt, hydrodynamic, velocity-bunching mechanisms of §2.3, using VV amplitude imagery as an alternative to the cross-polarimetric phase. Three complementary parameters were computed; The *azimuth cutoff* (λ_c) is obtained from the azimuth auto-covariance of detected intensity over an ocean patch, fitted to a Gaussian, and requires no MTF or inversion, since λ_c scales directly with the orbital-velocity spread of the sea surface and hence with wind sea and significant wave height; this makes it the most robust single-image sea-state parameter available and the natural first deliverable against the sea-state characterisation objective set for this strand of the internship.

$$C(\Delta x) \approx \exp\left(-\left(\frac{\Delta x}{\lambda_c}\right)^2\right) \tag{9}$$

The *2D image variance spectrum* is obtained by averaging windowed periodograms over multiple ocean sub-scenes, with peaks in $k = (k_{az}, k_{rg})$ giving the dominant swell wavelength $\lambda = 2\pi/|k|$ and propagation axis *(Eq. 5*'s spatial-domain counterpart). Because P is symmetric under $k - k$, this fixes the wave axis only modulo 180°.

$$P(k_{az}, k_{rg}) = \left|\mathrm{FFT}_{2D}\left(\Delta I/\bar{I}\right)\right|^2 \tag{10}$$

The ambiguity is resolved by the third parameter; *the sub-look cross-spectrum.* This is formed by splitting the synthetic aperture into two looks separated in slow time and forming the cross-spectrum between them. It yields an imaginary part that is antisymmetric in k. Its sign gives the true propagation direction. This is the same technique used by Sentinel-1 Wave Mode, and is particularly relevant because it is the only one of the three parameters that resolves direction rather than orientation.

2.5 Background of Future Work: Cryospheric Subsurface Applications

The penetration property discussed in §2.2 extends beyond the ocean's surface into the cryosphere. Over snow and ice, P-band appears to penetrate the dry-snow column largely unattenuated, so the dominant returned signal originates not from the air-to-snow interface but from buried interfaces of higher dielectric contrast, potentially firn-to-ice transitions, englacial layering, or the ice-bed interface itself. In principle this makes P-band polarimetry sensitive to grounding-line position and subglacial water bodies, high-value targets for ice-sheet mass-balance science that C-band and L-band SAR cannot directly image. An analogous argument applies over arid terrain, where the same penetration into dry, low-loss sand or soil has recently been shown to reveal buried palaeochannels [17] and near-surface hydrological structures invisible at shorter wavelengths. These applications were not investigated independently within this internship at time of writing, but are identified here as natural extensions of the same underlying physics. Future work is further described in §7 and outlines the progression of the internship. A general overview of how the scientific background relates to future work can be seen in the flowchart below.

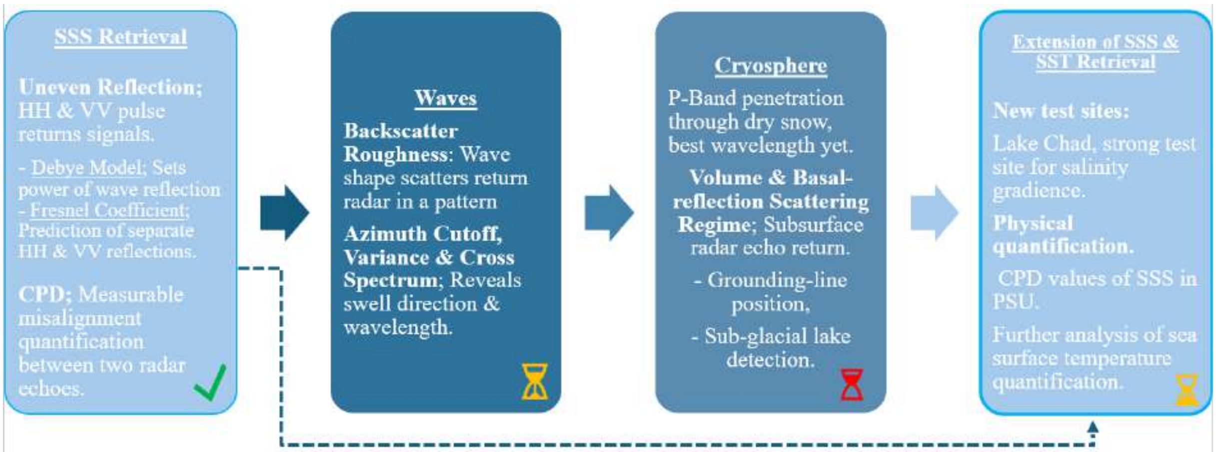


Figure 2: General basis of Biomass ocean and cryospheric subsurface applications, showing the strands explored during this internship

3 Methodology

3.1 Overview

This chapter outlines the overall methodology and tasks completed to achieve the objectives outlined within §1.3. Here, §3.2 describes the target ROI, reasoning of choices and the datasets used to test them. §3.4 describes the general experimental processing pipeline used in the SSS and SST retrieval, developed and tested with some improvements applied suited to each region. Table1 details a general overview of what differs between each ROI, including acquisition geometry, date of data acquisition, and relevant swath information, as the overall processing method itself does not change greatly.

The central question this chapter addresses is whether the physical chain established in §2, specifically *Eqs 4 & 6-8*, are not only theoretically viable, but also visually detectable in this data. A secondary, independent question, addressed in §1.3, is whether the same SLC products support sea-state characterisation via the tilt, hydrodynamic and velocity-bunching mechanisms of §2.

3.2 Regions of Interest

Four regions were selected for analysis, each chosen for a documented, sustained sea surface salinity contrast associated with freshwater discharge, and each cross-checked for Biomass Level-1a coverage using ESA MAAP Explorer[18] and the purpose built archive-scanning code found in the Appendix-C.3 before being used as a test case.

Table 1: Summary of ROI, acquisition details, and early reference salinity contrast.

Region	Track(s)	Bounding box (lon, lat)	Date(s)	Frame(s)/Product	SSS contrast (PSU)
Guiana Shelf (Demerara)	T036, T028	44-50 W, 6-9 N	(3-day int) 06,27-07-2026	F007-10, F154-155	20-36
Amazon River delta	T020, T021, T028	46-51 W, 2 S-5 N	23,26,29-07-2026	F001-5, F154-62, F304-10	8-37
Ganges-Meghna delta	T008, T015, T030	86-93 E, 19-24 N	21-11-2025, 03-08-2026	F014-17, F136-138	15-35
El Niño case study	T030-33, T037-40	175 W-95 W, 9 S-9 N	30-06-2026	F001-09, F303-310, F146-162	33-37

Guiana Shelf (Demerara Plume). The Guiana Shelf off Demerara receives the combined outflow of the Amazon plume transported alongshore together with local river discharge, and has a salinity contrast of approximately 20-36PSU. This was the most completely worked region and the one against which the processing pipeline below; §3.4, was first developed.

Amazon River Delta. Further south along coast, the Amazon River discharges the largest freshwater flux of any river system on Earth, producing a low-salinity plume that extends northwestward along the Guiana coast under the influence of the North Brazil Current, with surface salinity contrasts ranging from below 10PSU near the river mouth to open-ocean values above 36PSU within a few hundred kilometres offshore. This makes it the largest and best-documented salinity gradient available to the project, and a natural test case for validating retrieval performance across a wide dynamic range. However, as Biomass ocean coverage is still being actively acquired, this region remains largely uncovered by the mission so far.

Ganges-Meghna Delta (Bay of Bengal). The Ganges-Meghna system was selected as a geographically and hydrologically distinct test case from the Amazon/Guiana pair, as it's situated in a monsoon-driven delta. Validation for this region used SMAP L2B swath salinity alongside SMOS Level-3, since SMOS coverage was found to be poor this close to the delta coastline.

El Niño Case Study. A fourth region in the open ocean prompted mainly by a late July 2026 ESA report [19] on SMOS detection of the Pacific salinity shifts as the freshwater pool moved eastward with the onset of El Niño salinity. This stands out because this detection case is a rare phenomenon that, as SMOS and Biomass Mission Manager; Klaus Scipal put it, acts as "a fingerprint of the water cycle." Since SMOS and SMAP already resolve this at basin scale, the case instead tests whether Biomass's finer resolution adds anything at the local scale. In addition, being open ocean instead of coastal, offers a natural test site for the wave sea-state pipeline of the ocean methodology discussed above, alongside the salinity chain, letting both strands of this project run on the same scene.

3.3 Data access

Biomass Level-1a SLC products were retrieved from the ESA MAAP STAC catalogue [18] via the pystac client, authenticated through ESA's identity service: a 90-day offline refresh token, generated once via the MAAP portal [18], is exchanged for a short-lived access token attached to search and download requests.

Server-side pagination is capped at 100,000 records per query, and the product-type filter distinguishes standard acquisitions (S1_SCS__1S) from ocean calibration variants (S2_SCS__1S). Bulk download for a region uses the cell in §C.4 to query granules intersecting a bounding box and date range, filter by product type and frame availability, and download the four polarimetric GeoTIFF pairs (HH, HV, VH, VV) with each granule's KML footprint and annotation XML.

Reference salinity and temperature fields came from two further authenticated services: CMEMS products (GLORYS12 reanalysis, SMOS swaths, and OSTIA SST [20]) via the Copernicusmarine client, and SMAP L2B swath salinity from NASA PO.DAAC via Earthaccess [21]. ESA WorldCover [22] 10m land-cover tiles were downloaded once and cached locally, since the land mask is identical across acquisition dates for a given region.

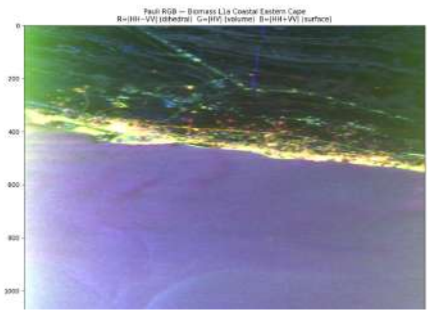
3.4 Common Processing Pipeline

Below outlines the multi-stage pipeline applied to all aforementioned ROI above.

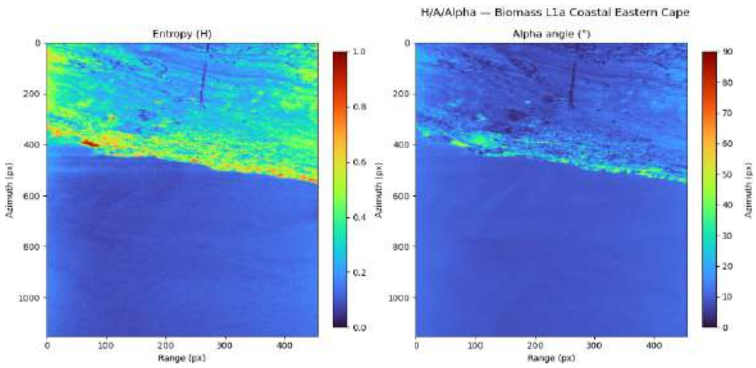
1. **Region and product identification.**
   Candidate regions were selected on the basis of documented salinity to freshwater plume contrast as discussed in §3.2. This step follows the sensitivity argument of §2.3, because the Fresnel/Debye chain *(Eqs. 6-8)* predicts a P-band CPD response of order $\sim$0.0013 dB/PSU in the polarimetric ratio, a genuinely detectable signal requires the largest available salinity gradients more than that found in typical open-ocean variability.

2. **Bulk download.** All available Level-1a Single-Look Complex (SLC) products for the selected region and dates were downloaded through the ESA MAAP STAC catalogue [18] using a bulk-download script in Appendix C.4. SLC files by choice over multi-looked or detected products are required because *Eq.3* depends on the coherent complex covariance term $C_{13} = \langle S_{HH} S_{VV}^* \rangle$; any product that has already discarded phase information can't support the CPD measurement.

3. **Baseline Product Visualisation.** For each granule of significance, plots were generated as a first-pass quality check, including multilooked intensity, Pauli RGB decomposition, all four polarisation visuals, and H/A/α decomposition (background and code in Appendix C.5). This verifies, before any salinity-specific processing, that the scattering regime over open water matches the first-order Bragg assumption underlying the forward model (§2): low entropy and α close to 0° over ocean pixels signal that S_{HH} and S_{VV} are appropriate, while elevated entropy or α shifted toward volume scattering would indicate roughness or non-Bragg contamination requiring separate treatment before Eq. 4 could be trusted.



(a) Pauli Example Image, See §C.5.3 for interpretation details.



(b) Entropy & $\alpha°$ §C.5.4 Example Images

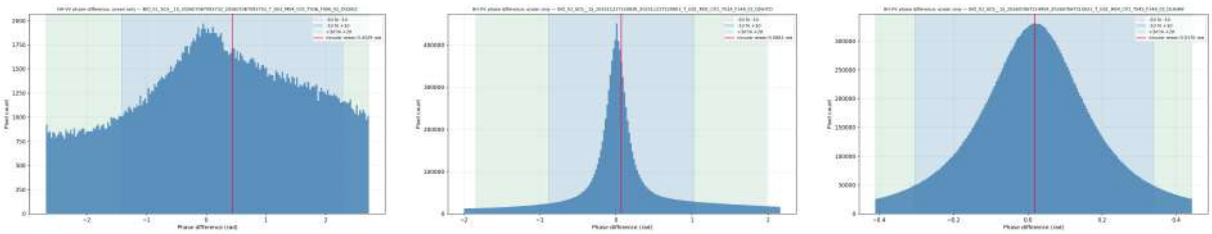
Figure 3: Example plots for baseline product visualisation

4. **Land/Coastal masking.** Where a scene included coastline or islands, a mask was built using the dedicated ESA WorldCover mask classifications sampled at per-pixel coordinates derived from the KML footprint corners. This is not just a cosmetic step: land backscatter at P-band is strongly volume- and double-bounce-dominated over Bragg Scattering, so any land contribution surviving into a covariance block would bias C_{13} and hence β *(Eq.4)* toward a value with no relationship to *Eqs. 6-8*, silently corrupting the salinity inversion as opposed to only adding noise to it.

5. **HH-VV phase difference.** The copolar phase difference $\varphi_{HH} - \varphi_{VV} = \arg(\beta)$ was computed per pixel; This was sought as a direct parameter of *Eq.* (4), and displayed using the cyclic `gist_rainbow` colour map. A cyclic map is a theoretical necessity more than a nice visual choice: because φ is defined only modulo 2π, a

linear (non-cyclic) colour scale would introduce a spurious discontinuity at the wrap point that has no physical meaning and could be mistaken for a real salinity front.

6. **Statistical conditioning.** The phase distribution within each block was visually inspected via a Gaussian histogram to identify outliers inconsistent with the expected circular-normal form, then condensed to its circular mean and $\pm 1\sigma$ range using $\bar{\varphi} = \arg\left(\langle e^{i\varphi} \rangle\right)$ and the associated circular spread. This conditioning step is required as phase is circular, as such; an arithmetic mean and standard deviation computed directly on wrapped φ values would be biased and, near the wrap boundary, virtually meaningless. The resulting $\pm 1\sigma$ range is what is subsequently compared against the speckle-limited precision $\sigma_\varphi = \sqrt{(1-\gamma^2)/(2N\gamma^2)}$ to establish whether an observed deviation is distinguishable from the noise floor before any salinity claim is made.



(a) Biased Gaussian (outliers) (b) Low deviation spread (c) Corrected spread

Figure 4: Contrast visual examples of statistical conditioning histograms.

7. **Forming Path Mosaic.** Individual frame-level phase-difference products belonging to the same track and date were placed into a single stitched product path mosaic, with frame boundaries preserved for traceability. This directly supports the frame-to-frame differencing as adjacent frames from the same datatake share instrumental state, orbit geometry and ionospheric path, any constant offset between them is instrumental rather than physical, and creating a product path mosaic with visible frame boundaries makes any such discontinuity immediately inspectable as opposed to hidden inside a single averaged product.

8. **Quantification.** Each mosaic, and separately each constituent frame, was reduced to a circular mean phase value $\bar{\varphi}$ and represented on a shared colour scale, allowing direct visual and quantitative comparison of phase magnitude across products and regions. Thereafter $\bar{\varphi}$ is compared against the CPD predicted by *4* for the reference salinity at that location, converting an inverted apparent dielectric parameter back to an implied SSS.

9. **External validation.** The resulting phase-based salinity signal was compared against co-located SMOS Level-3 and/or SMAP L2B swath salinity for the same scene and acquisition window, and against GLORYS12 reanalysis, assessing correlation

between the phase-derived and independently measured SSS. This is the ultimate test that SSS, as acquired using permittivity differences, predicted by the Fresnel CPD *(Eqs. 4, 6-8)* is not only theoretically real but detectable in actual Biomass data. Additionally where applicable, Sentinel-3 datasets are used to support this processing as an SST visual inspection tool for independent temperature validation.

4 Results

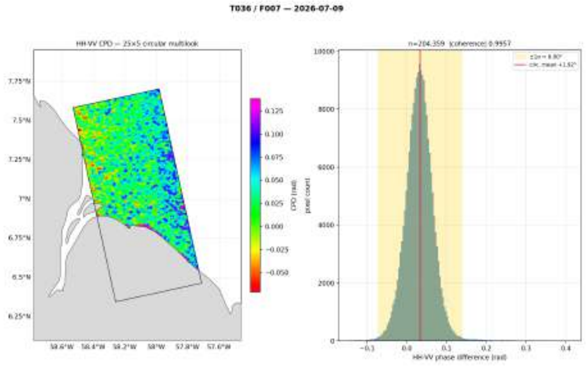
This section presents results from the four ROIs described in §3.2, following the methodology in §3.1. Across all four regions, HH-VV copolar phase difference resolves spatially coherent contrast between freshwater-influenced and open-ocean water masses, tracking independently with SMOS/SMAP salinity rather than arising from speckle, instrumental offset, or land contamination. The strength of this correspondence varies by region, as detailed below.

4.1 Guiana Shelf (Demerara Plume)

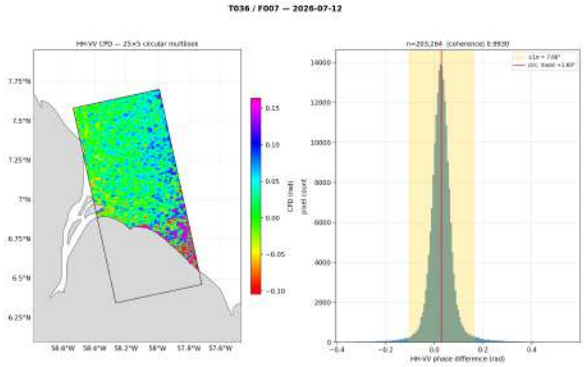
The Demerara plume was the most extensively sampled ROI of the three, with six Biomass acquisitions over Track 036/Frame 007 spanning 9th until 24th of July 2026 (Figure 5). Product coherence remained consistently high across all six passes (0.9919–0.9968), and the circular mean phase stayed within a narrow $+1.69°$ to $+2.20°$ band with no systematic drift over the 15-day window, suggesting that instrumental and/or ionospheric contributions to CPD are small and repeatable at this site.

Comparison against coincident SMAP and SMOS salinity (Figure 6) confirms the region carries salinity contrast against SMAP. SMAP L2B swath points on 6/07 span 20.05-35.86 PSU, while the narrower 32-34.5 PSU range in the SMOS daily composites on 12/07 and 24/07 is consistent with coverage-limited, closer-to-plume-core sampling. The same-date SMOS L3 vs. L2Q comparison on 12/07 (Figure 6b,d) shows agreement in overall range (33.06-34.44 vs. 33.14-35.38 PSU) but differing finer spatial structure, worth noting if future work selects a single reference product for regression.

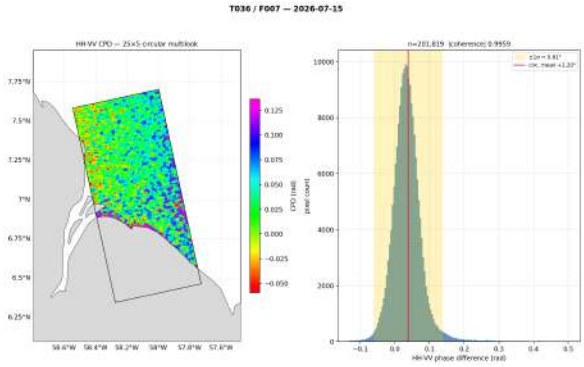
Taken together, the Guiana Shelf results support CPD's stability and repeatability at this coherence level as a candidate salinity indicator, but the analysis stops at that qualitative stage: no CPD-salinity regression has been completed here. §7 sets out the planned task to turn this repeat-pass stability into a quantified relationship.



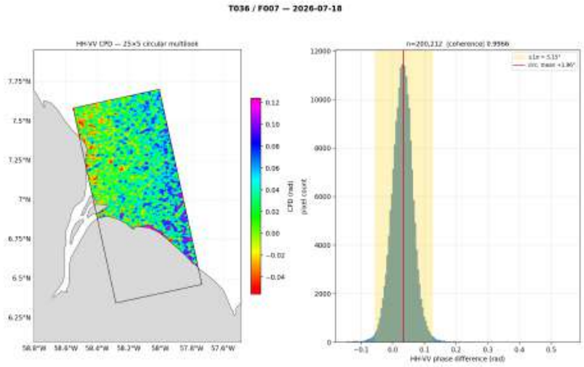
(a) 9/07 (coherence 0.9957, c.mean +1.92°)



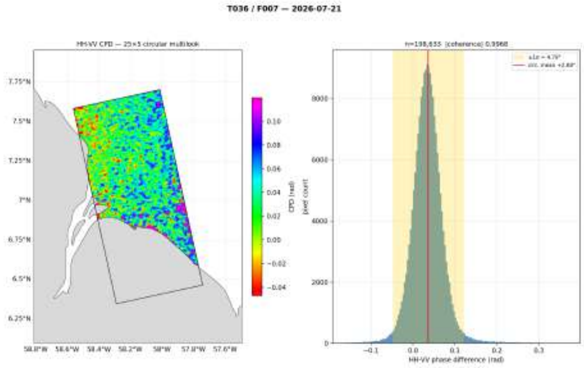
(b) 12/07 (coherence 0.9930, c.mean +1.69°).



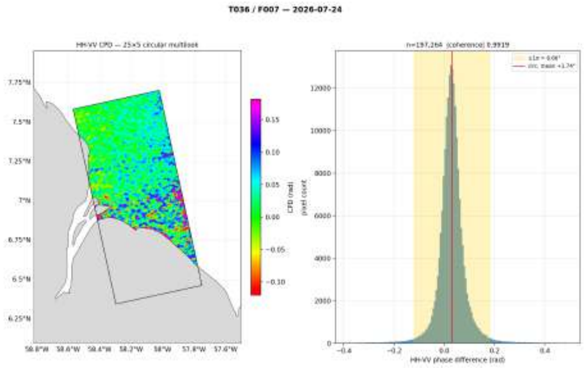
(c) 15/07 (coherence 0.9959, c.mean +2.20°).



(d) 18/7 (coherence 0.9966, c. mean +1.96°).

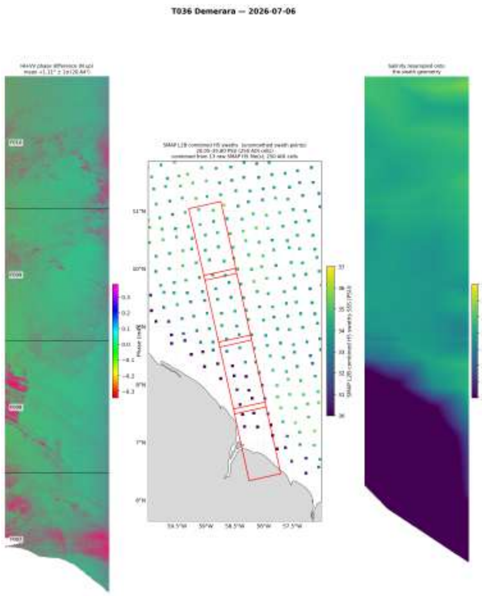


(e) 21/07 (coherence 0.9968, c. mean +2.08°).

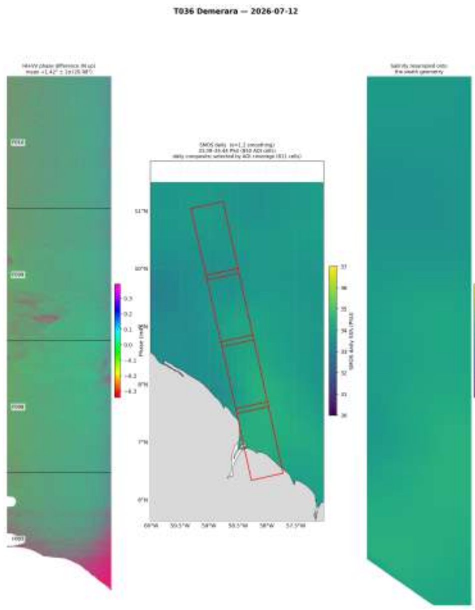


(f) 24/7 (coherence 0.9919, c. mean +1.74°).

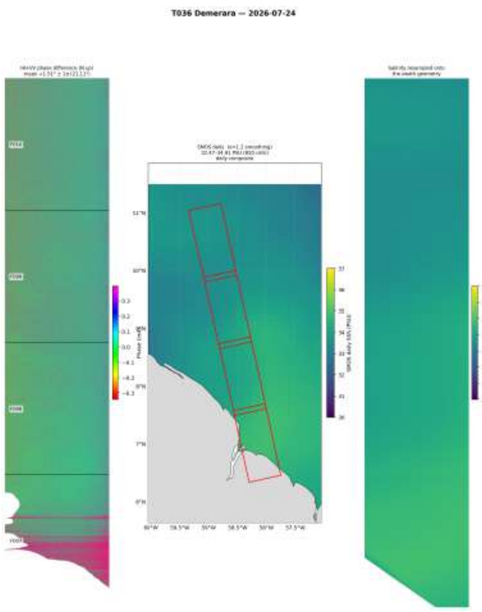
Figure 5: Four repeat-pass HH–VV CPD acquisitions over the Demerara plume (Track 036, Frame 007), spanning 9th July - 24th July 2026. Each panel shows the 25×5 circularly multilooked CPD map (left) alongside the corresponding pixel-count histogram (right), with product coherence, circular mean phase, and ±1 phase spread annotated.



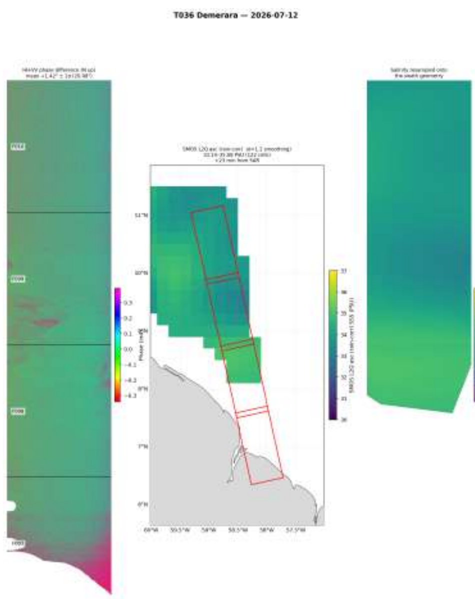
(a) 6/07 SMAP L2B combined half-orbit swaths (unsmoothed swath points, 20.05–35.86PSU, 1,153,428 cells)



(b) 12/07 SMOS daily composite (L3, 1.2 smoothing, 33.06–34.44PSU, coverage-selected daily composite).



(c) 24/07 SMOS daily composite (L3, 1.2 smoothing, 32.47–34.01PSU).



(d) 12/07 SMOS L2Q swath (near-real-time, 1.2 smoothing, 33.14–35.38 PSU, 22km from SSS), shown for direct comparison against the daily composite in (b) on the same acquisition date.

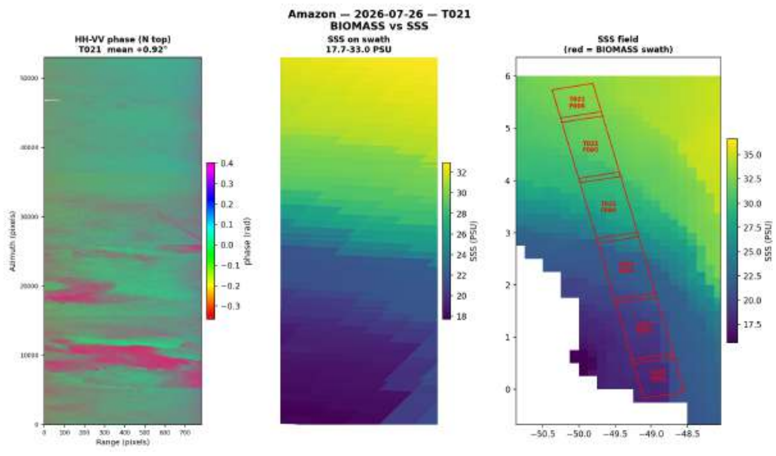
Figure 6: Biomass HH-VV CPD (left) alongside coincident satellite sea surface salinity, shown both in native product geometry (centre) and resampled onto the Biomass swath geometry (right), for the Demerara plume (Track 036). Panels illustrate the salinity resampling step across different source products and acquisition dates, including a same-date comparison of SMOS product types in (b) and (d).

4.2 Amazon River Delta

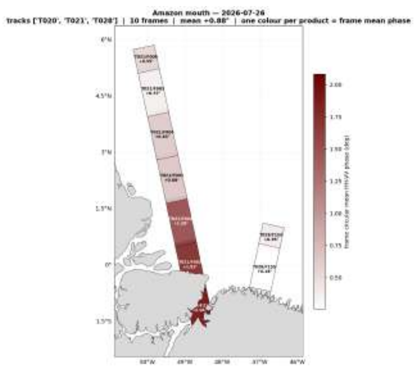
The Amazon delta was sampled across three tracks (T020, T021, T028) on three dates (23, 26, and 29 July), giving 9-10 frames per date once stitched into path mosaics (Figure 8). Track-level circular mean phase varied more here than in Guiana (+1.41° on 23/07, +0.88° on 26/07, +1.09° on 29/07), and coherence, while still high (0.82-0.94 across individual frames), was noticeably more variable frame-to-frame than the Guiana repeat-pass series, which is unsurprising given the Amazon mouth's more dynamic environment: stronger current shear, higher sediment load, and a more spatially compressed salinity front than the broader Demerara shelf.

The frame-level breakdown of T021 on 26/07 (Figure 7a) is the most direct look at CPD-salinity co-variation obtained in this project. Splitting the six-frame along-track sequence into a northern group (F001-F003) and southern group (F004-F006) shows salinity increasing from roughly 19 PSU near the river mouth (F001) to over 32 PSU offshore (F006), while mean phase broadly decreases across the sequence, consistent in sign with the expected dielectric response, though the relationship is not completely clean, as some elements of the upper path appear uncorrelated.

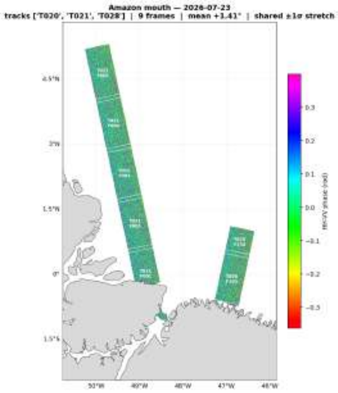
As with Guiana, this correspondence is presented as a visually and directionally consistent pattern, not a validated quantitative relationship. The steep along-track salinity gradient here (19 to 32 PSU over roughly 3.5° of latitude) strengthens the case that physical quantification is a worthwhile next step, since it would give real statistical results.



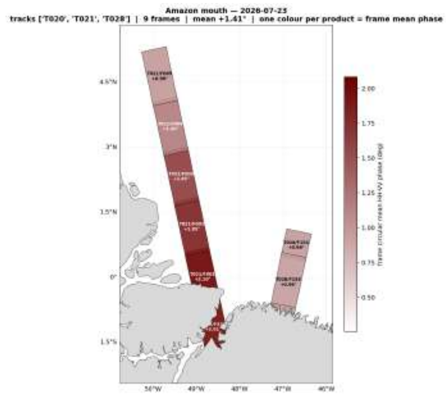
(a) HH–VV phase, salinity, and geographic context for T021 on 26/07. L-R: HH–VV phase strip with frame boxes, SSS resampled onto the swath geometry, and the SMOS/CMEMS SSS field with the Biomass swath overlaid (red).



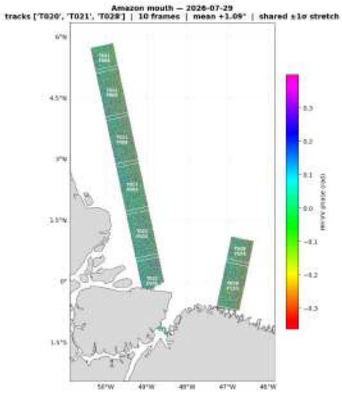
(b) 26/07, red-scale representation for T028 (10 frames, frame-level circular mean phase, mean +0.88°).



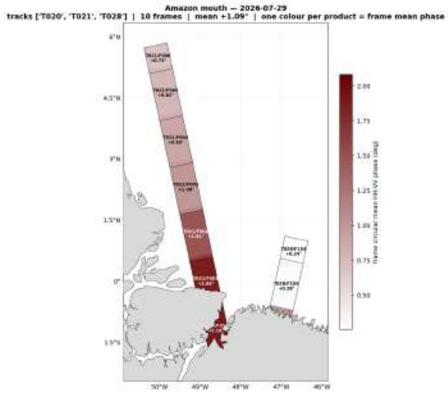
(a) 23/07, rainbow CPD scale (9 frames, tracks T020/T021/T028, mean +1.41°, shared ±1 stretch).



(b) 23/07, red-scale representation (one colour per product, frame-level circular mean phase).



(c) 29/07, rainbow CPD scale (10 frames, mean +1.09°, shared ±1 stretch).



(d) 29/07, red-scale representation (frame-level circular mean phase).

Figure 8: Stitched HH-VV phase-difference path mosaics for the Amazon delta on 23/07 and 29/07, shown across the rainbow (left) and red-scale (right) colour representations. Formed by combining individual frame-level products from the same track and date with frame boundaries preserved for traceability.

4.3 Ganges-Meghna Delta

The Ganges-Meghna delta was sampled on 03/08 across two tracks, T008 (3 frames) and T015 (4 frames) (Figure 9), plus single-frame detail for T008/F136 (Figure 10), which resolved a mean phase of $-1.73°$ at coherence 0.924 over 33% ocean coverage. This region shows sign-reversed phase relative to Guiana and Amazon on several frames, which was expected, since the Bay of Bengal plume sits within a different wind and current regime and φ_{CPD} sign depends on dielectric contrast rather than plume identity, but the reversal means the Ganges-Meghna result should not be pooled with the other two regions without first checking whether it reflects real physics or a residual instrumental offset specific to this track geometry.

Validation here is markedly weaker than at the other sites: although SMAP coverage was available for the two multi-frame tracks, no coincident salinity product could be located for the T008/F136 detail granule, leaving that acquisition without independent check. Given the Bay of Bengal's dense coastal geometry and correspondingly patchy SMOS/SMAP swath coverage, this is a limitation of the validation datasets rather than a processing failure, but it leaves Ganges-Meghna with the least complete evidentiary basis of the three regions, one that could be expanded in future work. The Ganges-Meghna results show that the pipeline generalises beyond the Atlantic sites it was developed on, rather than serving as evidence for the CPD-salinity relationship. Extending validation coverage and resolving the phase-sign question against a larger frame sample are needed before this region can meaningfully contribute to the sensitivity coefficient proposed in §7.

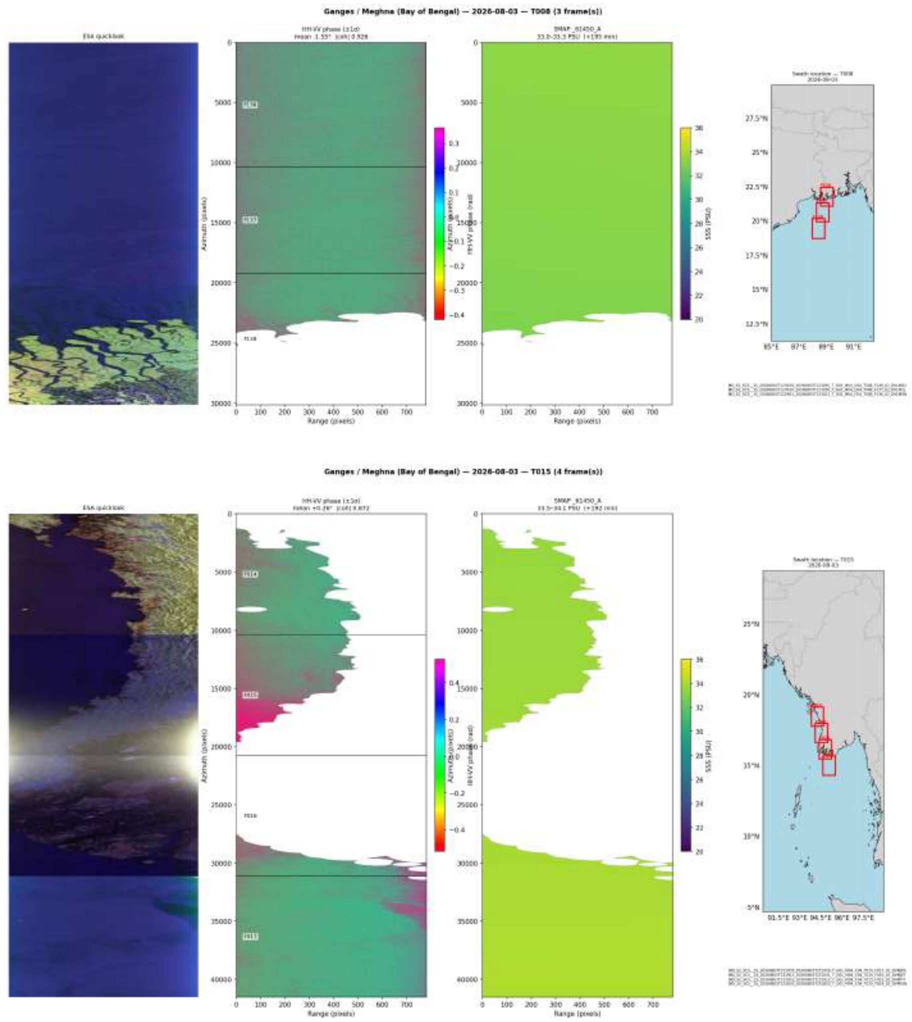


Figure 9: Output quicklook, HH–VV phase difference, SMAP salinity, and georeferencing for the Ganges-Meghna delta (Bay of Bengal), 03/08. From left to right: Pauli Scattered quicklook image, stacked HH–VV CPD frames with per-frame circular mean and coherence, coincident SMAP L3 SSS resampled onto the swath, and a location map showing the track footprint (red) relative to the delta. Top: track T008 (3 frames). Bottom: track T015 (4 frames).

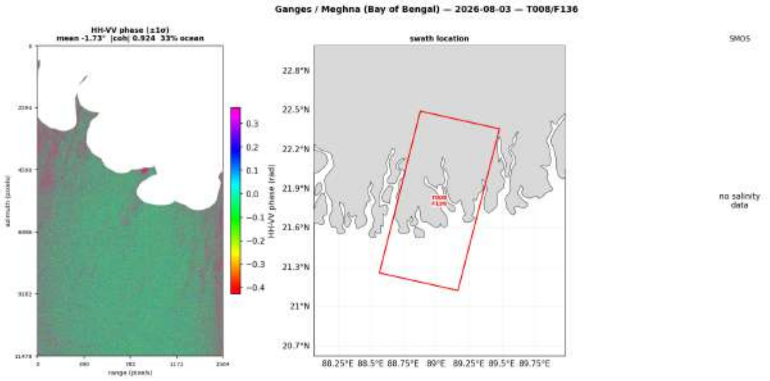


Figure 10: HH–VV CPD for T008/F136 over the Ganges-Meghna delta, 03/08. Left: 25×5 circularly multilooked CPD map, Right: swath location relative to the delta coastline. No coincident salinity product was available for this granule.

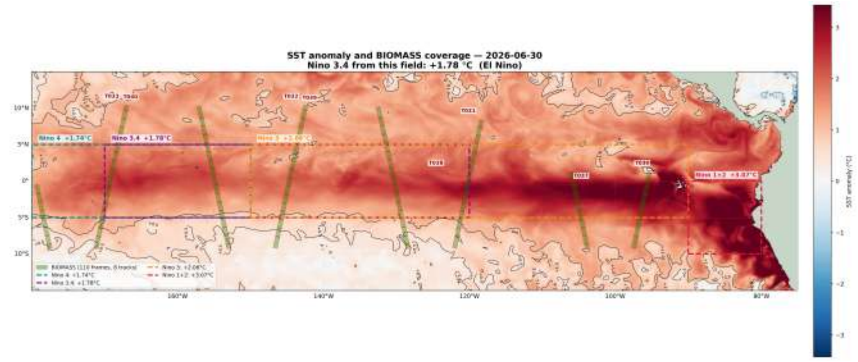
4.4 River Plume Cross-Region Summary

The three regions differ more in evidentiary strength than in method. Guiana makes the strongest case: six repeat passes at consistently high coherence (0.99+) and a tight phase band (+1.69° to +2.20°), against a clear 20-36 PSU salinity contrast. The Amazon delta is directionally consistent but noisier coherence 0.82-0.94, mean phase +0.88° to +1.41° and gives the clearest single result of the project: falling phase against rising salinity along T021 (19-32 PSU), undercut by an unexplained outlier at F005. Ganges-Meghna is weakest: phase sign reversed on several frames, and one granule (T008/F136) had no salinity validation at all; its main contribution is confirming the pipeline generalises to a third basin, not standalone support for the CPD-salinity link.

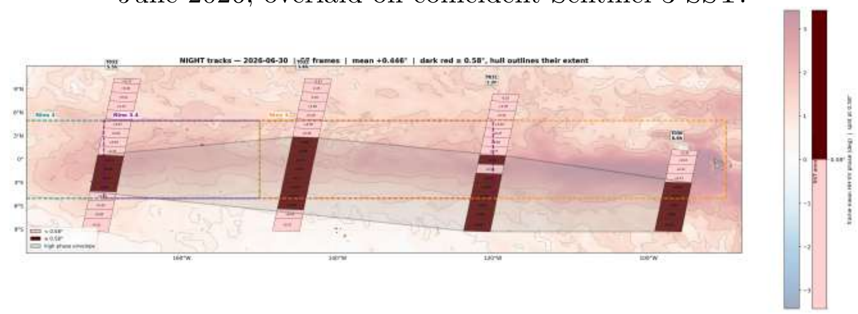
None of these results are quantified: every correspondence reported in §4.1-4.3 is visual and directional, not regressed. §7 sets out the empirical CPD-salinity regression needed to turn this into an actual coefficient, prioritising Guiana and the Amazon T021 gradient as the best-conditioned transects.

4.5 El Niño - Interest Based Case Study

This case study tested whether Biomass's finer spatial resolution could resolve structure within the freshwater pool at a scale finer than SMOS' or SMAP's basin-scale products. Figure 12 shows the full set of processed swaths (top) alongside SMOS validation mosaics. The eastern CPD panels show high-phase regions aligning well with areas of high SSS contrast in the corresponding SMOS panel below, a correlation to be quantified in future work. Critically, Figure 11a shows that regions of anomalous mean product phase response correlate spatially with the known El Niño freshwater 'tongue' as independently mapped by Sentinel-3 temperature, rather than appearing as isolated or random bias across the swath set. This is an encouraging result as it may suggest Biomass's resolution adds value over the coarser basin-scale SMOS product that originally motivated this case study.



(a) Biomass swath coverage within the El Niño study region, 30 June 2026, overlaid on coincident Sentinel-3 SST.



(b) Mean CPD within the highlighted region spatially corresponds to the documented El Niño SSS "tongue" in the southern Pacific.

Figure 11: Biomass product coverage for the El Niño case study, with Sentinel-3 SST overlay. (b) shows CPD-SSS correspondence consistent with validation method [§ 3.1]

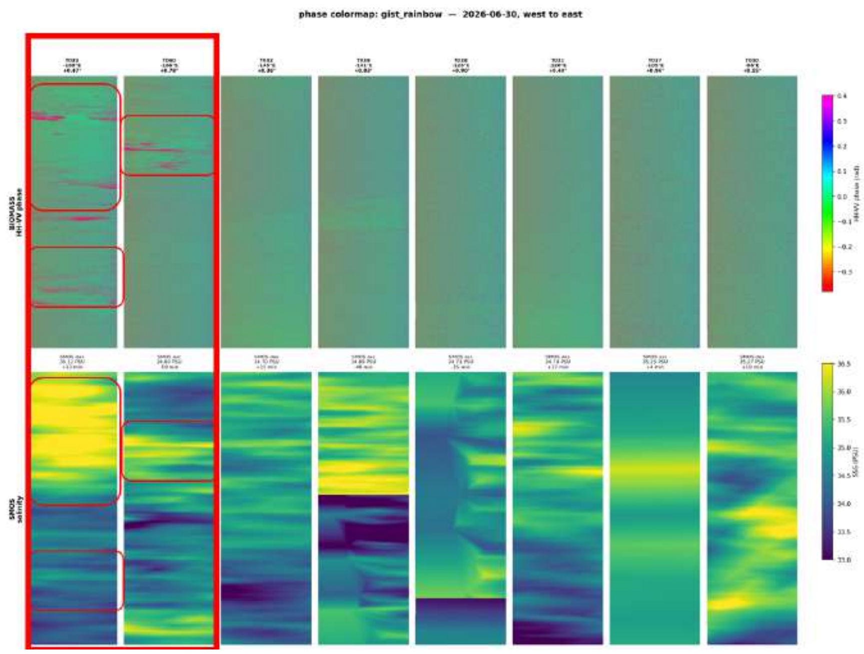


Figure 12: Full processed swath mosaics (top) compared with coincident SMOS SSS validation mosaics (bottom). In the most Eastern CPD panels, regions of elevated phase align spatially with areas of high SSS contrast in the corresponding SMOS panels.

5 Discussion

5.1 Interpretation of Results

The plume region results in §4 support the central claim motivating this project's core question of whether CPD can visually show SSS contrast. This is validated as theoretically sound and visually detectable in the data. Across the distinct ROI, the retrieved phase generally distinguished freshwater-influenced ocean regions from documented saltwater regions, and this distinction was independently corroborated by SMOS and/or SMAP salinity fields in every case where reference coverage was adequate. (§4.1-§4.3). This is a useful result given the predicted P-band polarimetric sensitivity is small, the signal was expected to sit close to the sensor's speckle-limited detection floor, and the fact that a consistent, somewhat spatially coherent and physically plausible pattern emerged over just noise is itself informative about the viability of P-band CPD salinity sensing as a concept.

The El Niño case study (§4.5) is additionally scientifically intriguing, because it was the region with the least directly comparable prior Biomass work, and completed based only on an interest basis, yet showed promising correlation with the SMOS and Sentinel-3 validation. Showing regions of higher phase response loosely corresponded spatially with the SMOS/SMAP-mapped freshwater tongue, rather than appearing as random scatters across the swath set, suggesting Biomass's resolution may offer some added value in this test case. This is a finding that, if it holds up under the more rigorous validation, would extend Biomass's demonstrated utility beyond the coastal-plume regime into open-ocean grand-scale salinity structures.

5.2 Limitations

Swath coverage and plume geometry. Biomass acquisitions are planned around the mission's primary forestry objective rather than systematic ocean coverage (§3.2), ocean swaths are new and opportunistic, while plume-region swaths are rare. In several regions, the available swath geometry clipped just beyond the true plume mouth rather than cutting through the region of sharpest salinity gradient, meaning the strongest possible signal was often not the one actually sampled. This is a coverage problem rather than a processing one, and should improve over the mission's lifetime as the acquisition archive grows. Figure 13 within the Appendix shows current available swath locations using the dedicated tool of Appendix C.3. It should be noted that only the red (coastal) and some blue (open ocean) products were usable for this project's purpose. There is also no coverage over Europe or the United States, excluding suitable plume cases such as the Western Iberia Buoyant Plume or the Mississippi River Plume, the latter with a dramatic freshwater-saltwater contrast into the Gulf of Mexico. Both are well documented by satellite and in-situ data, making them ideal candidates for future study.

Mission maturity. Biomass launched in mid-2025 (§1) and was barely a year into operations at the time of writing, early in its planned five-year lifetime. Both the acquisition archive and the broader ecosystem of processing tools, calibration refinements, and validation studies for P-band ocean applications are consequently far less mature than for long-established missions like SMOS or Sentinel-1. Several confounder-handling choices in this project (§3.4) are scene-averaged rather than pixel-level, reflecting what was practical within the internship timeframe, not the ceiling on what P-band salinity retrieval can ultimately achieve as the mission and its supporting tools mature.

Internship timeframe. As noted in §7, this report covers only the early to intermediate phases of a six-month internship, see Appendix-A.1 for full outline of internship time progression. The four-region, multi-scene result set presented here represents what was achievable within that window; the improvements and extensions outlined in §7.1-§7.2 are work that was scoped but not yet completed at time of writing. The results in this chapter should accordingly be read as a demonstration of feasibility and a foundation for the remainder of that work, more than a finished, fully hardened retrieval product.

Roughness/wave coupling. As discussed in §2.4, the SSS and wave strands of this project are not fully independent: surface roughness affects both the Fresnel-phase salinity and Bragg-scattering wave observables. As the wave-imaging strand was not carried far enough yet in time to produce a robust, independent roughness correction, the correction applied to the results so far remain a scene-averaged approximation rather than a fully decoupled treatment, which is an actual source of residual uncertainty in the visual values.

Geolocation and orientation handling. A further issue was found in how the ocean-masking pipeline assigned array orientation from each scene's KML footprint corners.[1] The original logic assumed a fixed corner orientation, which held for ascending passes but silently mirrored the scene for descending passes, since Biomass alternates between the two. This produced no error, only a mask misaligned with the true coastline, and was only caught by comparing a descending-pass mask against its Pauli RGB quicklook. The fix was to read `orbitPass` directly from each granule's annotation XML rather than assume a fixed orientation. All four regions in this report were manually verified as correctly oriented after the fix, though this remains a class of error worth auditing systematically rather than catching case-by-case.

[1]Included here for completeness rather than as a limitation on the results presented: the issue was caught and fixed during development, and did not affect the final results shown.

6 Conclusion

6.1 Summary of Achievements

Set against the three objectives stated in §1.3, this internship achieved good but uneven progress. The first objective, evaluating Biomass P-band sensitivity to sea surface salinity and, secondarily, to wave characteristics and sea-state, was an overall success, with work still to be completed: a full, re-implementable processing pipeline was developed and applied across three plume regions and an exploratory El Niño case study, producing consistent, high-coherence CPD retrievals validated against SMAP, SMOS, and CMEMS reference products. The wave and sea-state strand was scoped and partially implemented but not carried to the same depth within the available timeframe. The second objective, cryospheric observation capability, remained at the background and early-implementation stage (§2.5) rather than reaching full implementation, a scope decision made early in the internship to give the oceanographic strand the depth it needed, documented here rather than left unaddressed. The third objective, identifying broader Earth system science opportunities for Biomass beyond its forestry mission, is addressed throughout this report's framing and is the connecting thread between the ocean results obtained and the cryospheric extension proposed in §7.2.

6.2 Key Takeaways

- HH-VV CPD is retrievable from Biomass data at high coherence (0.82–0.997 across all products processed), and is stable enough, at Guiana Shelf in particular, to support a repeat-pass series with a narrow phase band over 15 days. This is itself informative: predicted P-band salinity sensitivity is small and expected to sit near the sensor's speckle-limited detection floor, so a consistent, physically plausible signal above noise speaks to the viability of the underlying concept before any regression is attempted.

- Across all three plume regions, CPD is directionally consistent with independently measured salinity contrast where validation coverage was adequate, most clearly at Guiana Shelf and along the Amazon T021 transect. This correspondence is visual and qualitative, not yet quantified, and its strength was not uniform: it degraded where swath coverage clipped short of the sharpest salinity gradient, where validation products were missing or incomplete (Ganges-Meghna), and where isolated frame-level anomalies remain unexplained. These are reported as open questions rather than smoothed into an averaged result.

- The largest practical constraint was swath coverage, not signal physics: Biomass acquisitions are planned around the mission's forestry objective, so ocean-crossing swaths over plume regions are opportunistic and do not always sample the sharpest part of a given salinity gradient. This should improve as the acquisition archive grows over the mission's remaining lifetime.

- No empirical CPD-salinity sensitivity coefficient has yet been derived; this project establishes the qualitative case for pursuing that regression and identifies Guiana and Amazon T021 as the best-conditioned transects for doing so, as set out in §7.2.
- Working with a newly-launched, still-maturing mission meant building processing tools and validation practices largely from first principles, without the established community tooling available for longer-running missions like Sentinel-1 or SMOS. That process, from geolocation/orientation handling to ocean masking to salinity co-registration, was as central to this internship's learning outcomes as the salinity results themselves, and leaves a considerably more auditable and extensible pipeline than existed at the project's outset.

7 Future Work

As this report covers only the work completed to date, a substantial amount of work remains planned, falling into two categories: improvements that consolidate and harden the existing CPD-SSS analysis, and extensions that broaden the investigation into new observables, test sites, and the cryospheric track.

7.1 Improvements

- **Confounder integration and pixel-level wind masking:** Planned work will replace scene-averaged wind corrections with hourly ECMWF wind and sea-state fields [23] co-located at pixel level, masking wind-contaminated regions rather than treating scenes as uniformly clean.
- **Ionospheric phase screen tracking:** Future processing will track per-product ionospheric correction quality flags systematically across the acquisition archive, so scenes with poorly corrected contributions can be excluded or down-weighted.
- **Consistent land masking across all scenes:** The land-masking pipeline developed for the Amazon delta scenes will be applied uniformly to all study scenes, including earlier South Atlantic acquisitions initially treated as pure open ocean.
- **Statistical robustness of phase estimates:** Further work will formalise the multilooking and averaging strategy, using the coherence-derived circular statistics of Section 2 to attach confidence intervals to every retrieved CPD value, alongside outstanding refinements to the polarimetric optimisation routines.

7.2 Extensions

- **Quantified CPD-SSS sensitivity coefficient:** This report establishes the qualitative CPD-plume correspondence but stops short of an empirical sensitivity number. Future work will regress CPD directly against coincident SMAP/reanalysis salinity across the validated plume transects, closing the gap between the theoretical Fresnel-derived prediction and a measurement-grounded coefficient.

- **CPD validation over the North Brazil Current:** The NBC retroflection offers a large, well-documented offshore salinity gradient from the Amazon-Orinoco plume, the strongest available open-ocean test case, and lies directly between the Amazon and Demerara sites. Planned work will compare Biomass CPD mosaics against coincident SMAP and reanalysis salinity [9] here once further coverage becomes available.

- **Controlled lake test site:** Inland water bodies with strong internal salinity gradients reduce the roughness and current confounders of the open ocean. Lake Chad is the preferred candidate given its position within Biomass coverage, with the Caspian Sea and Black Sea as alternatives.

- **Wave spectrum inversion:** Beyond salinity, Biomass's high native resolution invites cross-spectral analysis to extract swell direction and dominant wavelength, extending the mission's utility into sea-state characterisation [8].

- **Cryospheric track:** The internship's second objective remains to be executed: applying the same polarimetric and phase-based toolset to polar acquisitions. P-band's expected penetration through the dry-snow column, and the resulting shift from surface to volume-and-basal-reflection scattering, motivates prioritising grounding-line position and subglacial water bodies, both high-value targets inaccessible to C-band and L-band SAR.

- **Covariance-domain retrieval at P-band:** The inversion framework of Farahat and Hussein [16], demonstrated at L- and C-band, will be implemented against Biomass data as a complementary retrieval pathway to CPD, exploiting the cleaner separation of seawater ε-σ characteristic curves expected at lower frequency.

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A Internship Documentation

A.1 Work Packages & Internship Progression

The internship spans six months in total. Although this report provides a detailed overview of the work undertaken, it covers only the period from May to late August and therefore reflects the early and intermediate phases of the project. As the internship was still ongoing at the time of writing, several work packages had not yet been fully completed and are presented according to their current stage of development. In addition to the core internship objectives, a large component of the placement involved additional participation in the 2026 ESA-DLR Polarimetric SAR Interferometry (PolInSAR) Training Course delivered for 2.5 hours, three days weekly from mid-April, prior to the official start of the internship until mid-July and provided comprehensive guidance in polarimetric and interferometric SAR theory, data processing techniques, and practical applications [24]. The knowledge gained directly supported the internship project by strengthening the theoretical understanding required for the analysis of Biomass P-band radar data and related Earth observation methodologies. More information on this course can be found here.

- WP01: Induction, Training and Earth observation Re-familiarisation

The first phase of the internship focused on familiarisation with the working environment at ESA ESTEC and developing a strong foundation in EO principles, data products, and processing workflows. During this period, time was allocated to understanding the Biomass mission alongside other ESA Earth Explorer missions, including SMOS and CryoSat, and expanding on existing knowledge of Copernicus Sentinel data applications for both terrestrial and marine applications. Practical experience was gained using processing software such as ESA SNAP and QGIS along with additionally internally available Earth Explorer products, allowing confidence to be developed in handling large EO datasets. This work package provided the technical foundation required for the remainder of the internship.

- WP02: Literature Review and Scientific Background

Following familiarisation, attention shifted towards understanding the scientific motivation behind the internship project. A comprehensive literature review was conducted on P-band radar remote sensing, highlighting differences from conventional SAR systems, and the advantages it provides for environmental monitoring. Additional research focused on electromagnetic wave interactions with water surfaces, dielectric properties, and the physical principles governing salinity and waveform retrieval from radar observations. This work package established the theoretical foundation for all subsequent modelling activities thereafter.

- WP03: Data Analysis and Model Development

By late May and throughout June, the internship progressed from theoretical research to practical implementation. The primary objective during this stage was to investigate Biomass data products and develop a methodology capable of relating radar backscatter measurements to water salinity, in addition to exploring its land capabilities, specifically in coastal and desert related ROI. Coding work commenced to process available EO datasets, implement the proposed algorithms, and investigate relationships between dielectric properties and observed radar signals. Continuous testing and validation were performed throughout development to evaluate model performance and identify areas requiring refinement.

- WP04: Model Refinement and Validation

This WP was completed during July. Here, focus shifted towards improving the robustness of the developed methodology. Initial work centered on ocean salinity and temperature estimations sought from experimental analysis of dielectric behaviour. This required modifications to the existing model and further validation using representative datasets. Throughout this phase, continuous improvements were made to increase the reliability and scientific accuracy of the developed approach.

- WP05: Results Analysis and Technical Reporting

Toward late August, the emphasis moved from development and investigation to analysing results and documenting the completed work. Model outputs were evaluated against the research objectives, and conclusions were drawn regarding the performance and limitations of the methodology. These findings formed the basis of the internship report.

- WP06: Professional Development

In addition to the primary research role, the author participated in several ESA activities, including the ESA Internal Gravity Assist Conference, testing and providing feedback on astronaut helmet concepts, lunar surface engineering applications, and the European CanSat initiative, which included CDF (Concurrent Design Facility) training from Massimo Bandecchi, founder of Europe's first dedicated CDF. The author also supported educational outreach introducing secondary school students across Europe to spacecraft systems engineering and mission design, was appointed Science Design Engineer with YPSat, ESA's voluntary initiative for young professionals, following a competitive recruitment process, and was most recently selected for a user-friendliness study of SPOT, ESA's robotic dog developed by Boston Dynamics, investigating how humans can supervise robots performing infrastructure assembly tasks relevant to future lunar or Martian missions.

These experiences complemented the technical internship work by strengthening communication, teamwork, systems engineering, and public engagement skills, and provided exposure to multidisciplinary collaboration within the European space sector.

B Supplementary Material

All cleaned code notebooks and key results are available in the dedicated GitHub Repository. The supplementary code included in this appendix is limited to the core cells directly relevant to the analyses presented in this report. Readers interested in the full workflow, additional analyses, and implementation details are referred to the dedicated internship repository.

C Supplementary Code & Statements of Use

This appendix presents the core code developed during the internship in support of the methodology in Section 3.4, including the Biomass archive-scanning tool (§C.3), the product granule downloader (§C.4), the baseline product visualisation pipeline (§C.5); and the HH-VV phase-difference salinity retrieval pipeline (§C.6). Listings are trimmed to the cells directly relevant to this report; full notebooks are available in the accompanying GitHub repository. The origin and development of this code, including where AI tools were used, is disclosed in C.1

C.1 Declaration of AI Use

It should be noted that AI assistance was used in noted cells within this supplementary section to varying degrees, mostly for presentation purposes. The Biomass Full-Catalog Swath Scanner tool (C.3) was independently designed and coded by the author, with AI support limited to troubleshooting and formatting. The downloader cells and the baseline product visualisation pipeline (C.5)(multilooking, Pauli decomposition, and H/A/α decomposition) are the author's own code, adapted from material provided during the weekly ESA/DLR Biomass training course [24] attended during this internship. For the salinity-retrieval and wave-imaging pipelines, the physical reasoning and mathematics (§2, §2.4), together with all plotting and visualisation code, are the author's own; the author specified the required logic and, in places, supplied pseudocode, with AI assistance used to implement this into working code, including the CPD inversion routine. This aligns with ESA IT onboarding guidance encouraging interns to use LUNA (ESA's internal AI assistant) and Microsoft Copilot for programming support where neccesary. An honest record of *some* of the AI-assisted prompts used is provided here.

C.2 Prerequisite - ESA MAAP Credentials

Access to ESA MAAP, where the Biomass data is stored, requires a verified set of credentials. Once acquired, these must be saved locally in a separate plain `eo_credentials.txt` file that sets `maap_offline_token`, `maap_client_id`, and `maap_client_secret`, obtained

from a MAAP account at https://explorer.maap.eo.esa.int/. These same credentials are used throughout Biomass' work, well beyond just the project scanner, download, and processing cells below. The offline token expires periodically after 90 days or after a stretch of inactivity, with no warning beforehand, so something that has run fine for weeks can suddenly fail on the very first cell with a 401 error. The fix is just generating a new token from the MAAP account and swapping it into `eo_credentials.txt`, but it's easy to forget in the new token generation.
As such, it's worth highlighting here as a natural introduction to the supplementary code.

```
maap_offline_token = 'eyJhbGciOiJIUzUxMiIsInR5cCIgOiAiSldUIi...'
maap_client_id = 'offline-token'
maap_client_secret = 'p1eL7uonXs6MDxtGbgKdPVRAmnGxHpVE'
```

Listing 1: Format of `eo_credentials.txt`

C.3 Biomass Full-Catalog Swath Scanner

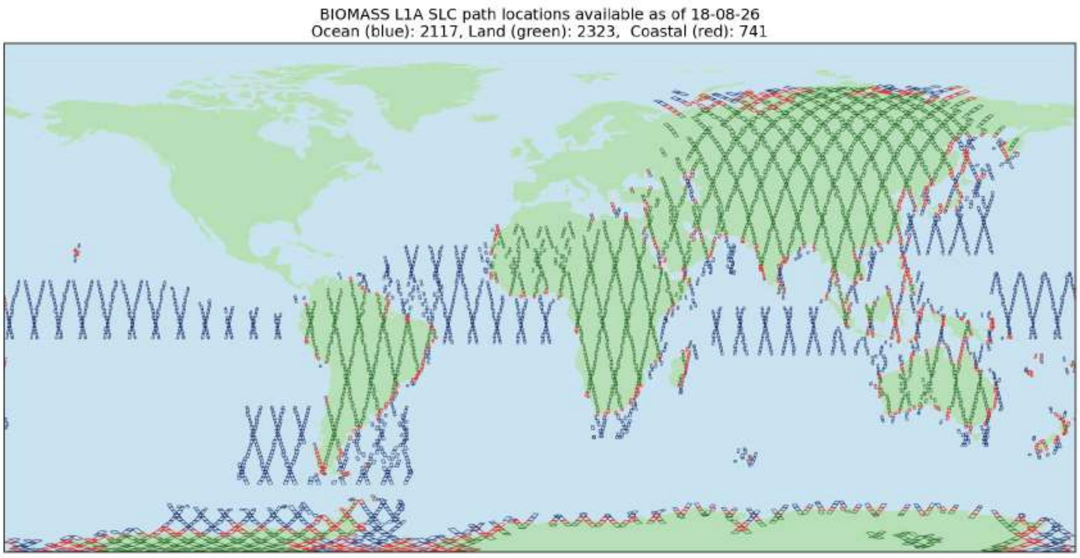


Figure 13: Available Biomass product locations during internship timeframe, note only blue (ocean) and red (coastal) products were used during this internship.

Biomass revisits the same swath locations repeatedly over its mission lifetime to support tomographic repeat-pass acquisitions, so its L1a SLC catalog holds over 200,000 granules at time of writing, but only a few thousand (roughly 2.3% of the full catalog) are genuinely unique swath footprints. Since validating sea surface salinity retrieval from Biomass P-band data first required knowing where the mission had actually imaged, rather than processing every individual acquisition, `QuickSwathScanner.ipynb` was written to scan the full ESA MAAP `BiomassLevel1a` STAC collection once and build a deduplicated catalog of those unique swaths, classifying each by its ocean, land, or coastal fraction using Natural Earth coastline data. A full scan takes anywhere from 8 to 11+ hours and is routinely interrupted by transient server errors, so the scanner checkpoints its progress and resumes automatically after any interruption. The code below is trimmed to the parts that actually matter methodologically: the configuration, the self-healing scan/deduplication loop, the discovery-curve check used to confirm the catalog was representative rather than a partial snapshot, and the resulting classified coverage map.

C.3.1 Imports and Path Configuration

```python
from pathlib import Path
SECRETS_PATH = Path.home() / 'eo_credentials.txt' # ESA EO credentials
BASE_DIR = Path.home() / 'STEP_BACK' / 'sidequest_check' # this is convenient paths for my setup but yours may differ...
COLLECTION_ID = "BiomassLevel1a"
KEYCLOAK_TOKEN_URL = "https://iam.maap.eo.esa.int/realms/esa-maap/protocol/openid-connect/token"
CATALOG_URL = "https://catalog.maap.eo.esa.int/catalogue/"

BATCH_SIZE = 500 # granules between checkpoint writes
MAX_OUTER_ATTEMPTS = 50 # kept high, the API drops out A LOT over a multi-hour scan
OUTER_RECONNECT_DELAY = 30 # seconds to wait before each reconnect attempt
OCEAN_THRESHOLD = 0.95 # ocean_fraction >= this -> "ocean"
LAND_ONLY_THRESHOLD = 0.05 # ocean_fraction <= this -> "land", else "coastal"
BASE_DIR.mkdir(parents=True, exist_ok=True)
STATE_FILE = BASE_DIR / "scan_state.json"
ALL_SWATHS_CSV = BASE_DIR / "all_swaths.csv"
ALL_ACQUISITIONS_CSV = BASE_DIR / "all_acquisitions.csv"
OUTPUT_DIR = BASE_DIR
```

Listing 2: Configuration: credential paths, MAAP API endpoints, scan parameters, and land/ocean classification thresholds for the swath scanner.

C.3.2 Setup / Authentication

Run once per session, reloads all already checkpointed so it's safe to re-run mid-scan.

```python
exec(SECRETS_PATH.read_text())  # maap_offline_token, maap_client_id, maap_client_secret.... needs to be set-up in advance.
import json, gc, time, requests
import pandas as pd
from shapely.geometry import shape
from shapely.ops import unary_union
from pystac_client import Client
from pystac_client.exceptions import APIError
```

```python
import cartopy.io.shapereader as shpreader
def get_access_token():
    resp = requests.post(KEYCLOAK_TOKEN_URL, data={
        "grant_type": "refresh_token",
        "client_id": maap_client_id,
        "client_secret": maap_client_secret,  # omitting this gives a 401 unauthorized_client so its definitely needed
        "refresh_token": maap_offline_token,
        "scope": "offline_access openid",
    })
    resp.raise_for_status()
    return resp.json()["access_token"]

catalog = Client.open(CATALOG_URL, headers={"Authorization": f"Bearer {get_access_token()}"})

# reload the checkpoint if it exists, otherwise rebuild it from the CSV (survives a deleted state file)
if STATE_FILE.exists():
    with open(STATE_FILE) as f:
        state = json.load(f)
elif ALL_ACQUISITIONS_CSV.exists():
    acq_df = pd.read_csv(ALL_ACQUISITIONS_CSV)
    acq_df["datetime"] = pd.to_datetime(acq_df["datetime"])
    max_dt = acq_df["datetime"].max()
    state = {
        "n_scanned": len(acq_df),
        "seen_grid_codes": sorted(acq_df["grid_code"].dropna().unique().tolist()),
        "last_datetime": max_dt.strftime("%Y-%m-%dT%H:%M:%S.%f")[:-3] + "Z",
        "boundary_ids": acq_df.loc[acq_df["datetime"] == max_dt, "id"].tolist(),
    }
    with open(STATE_FILE, "w") as f:
        json.dump(state, f)
else:
    state = {"n_scanned": 0}

# Natural Earth 50m land, lake-aware (lakes subtracted so inland seas like the Caspian read as water)
land_geoms = [shape(g) for g in shpreader.Reader(shpreader.natural_earth(
    resolution="50m", category="physical", name="land")).geometries() if g is not None]
lake_geoms = [shape(g) for g in shpreader.Reader(shpreader.natural_earth(
```

```
    resolution="50m", category="physical", name="lakes")).geometries()
   if g is not None]
land_union = unary_union(land_geoms).difference(unary_union(lake_geoms))
```

Listing 3: Setup: authentication, checkpoint reload, land mask.

C.3.3 Scan the Collection

This is the core important cell of the notebook: although its long-running, its also resumable and self-healing against the transient API errors a multi-hour scan is guaranteed to hit. Two bugs worth flagging:

1. Earlier version filtered `product:type` server-side, silently dropping valid `S2/S3` acquisitions with no error, fixed below via a client-side check instead.
2. `APIError` mid-page used to get misread as end-of-collection, silently truncating the scan after a single 502, fixed by only trusting a `StopIteration` from a fresh iterator.

```
class APIReconnectNeeded(Exception):
    """Raised on ANY APIError mid-scan, signals a fresh search() rather
   than
    retrying the now-unreliable iterator."""
    pass

def next_or_reconnect(it):
    try:
        return next(it)
    except StopIteration:
        raise  # a real StopIteration = genuine end of collection
    except APIError as e:
        raise APIReconnectNeeded(str(e)) from e

seen_grid_codes = set(state.get("seen_grid_codes", []))
write_header = not ALL_SWATHS_CSV.exists()
write_header_acq = not ALL_ACQUISITIONS_CSV.exists()
n_new = state["n_scanned"]
n_duplicate_skips = 0
n_wrong_product_type_skips = 0
last_seen_datetime = state.get("last_datetime")
boundary_ids = set(state.get("boundary_ids", []))  # ids already
   processed at the resume datetime, avoids double-counting/skipping
   ties

outer_attempt = 0
finished = False

while outer_attempt < MAX_OUTER_ATTEMPTS and not finished:
    outer_attempt += 1
```

```
    resume_datetime = state.get("last_datetime")
    search = catalog.search(
        collections=[COLLECTION_ID], method="GET",
        datetime=f"{resume_datetime}/.." if resume_datetime else None,
        sortby="+properties.datetime",  # ascending, so "resume from last datetime" is well-defined
        max_items=None,
    )
    item_iter = search.items()  # fresh iterator + fresh HTTP session, the whole point of reconnecting
    batch_rows, acquisitions_batch, page_error = [], [], False

    try:
        while True:
            try:
                item = next_or_reconnect(item_iter)
            except StopIteration:
                finished = True
                break
            except APIReconnectNeeded:
                page_error = True
                break

            item_dt = item.properties.get("datetime")
            if item_dt == resume_datetime and item.id in boundary_ids:
                continue  # already processed this one before the last interruption

            product_type = item.properties.get("product:type", "")
            if not product_type.endswith("SCS__1S"):  # client-side check, covers S1/S2/S3 without a server-side filter
                n_wrong_product_type_skips += 1
                continue

            n_new += 1
            if item_dt != last_seen_datetime:
                boundary_ids = set()
            last_seen_datetime = item_dt
            boundary_ids.add(item.id)

            grid_code = item.properties.get("grid:code")
            acquisitions_batch.append({
                "id": item.id, "grid_code": grid_code,
                "datatake_id": item.properties.get("eopf:datatake_id"),
                "absolute_orbit": item.properties.get("sat:absolute_orbit"),
```

```
                "datetime": item_dt, "orbit_state": item.properties.get("sat:orbit_state"),
                "product_type": product_type,
            })  # every acquisition logged, including repeats

            if grid_code is not None and grid_code in seen_grid_codes:
                n_duplicate_skips += 1  # already have this swath, skip the expensive geometry work
            else:
                if grid_code is not None:
                    seen_grid_codes.add(grid_code)
                try:
                    footprint = shape(item.geometry)
                    ocean_fraction = (1 - footprint.intersection(land_union).area / footprint.area
                                       ) if footprint.area > 0 else None
                except Exception:
                    ocean_fraction = None  # don't let one bad geometry take the whole scan down
                if ocean_fraction is not None:
                    batch_rows.append({
                        "id": item.id, "grid_code": grid_code, "geometry_wkt": footprint.wkt,
                        "ocean_fraction": ocean_fraction, "orbit_state": item.properties.get("sat:orbit_state"),
                        "datetime": item_dt, "product_type": product_type,
                    })

            if n_new % BATCH_SIZE == 0:  # checkpoint
                if batch_rows:
                    pd.DataFrame(batch_rows).to_csv(ALL_SWATHS_CSV, mode="a", header=write_header, index=False)
                    write_header, batch_rows = False, []
                if acquisitions_batch:
                    pd.DataFrame(acquisitions_batch).to_csv(ALL_ACQUISITIONS_CSV, mode="a", header=write_header_acq, index=False)
                    write_header_acq, acquisitions_batch = False, []
                state.update(n_scanned=n_new, seen_grid_codes=list(seen_grid_codes),
                             last_datetime=last_seen_datetime, boundary_ids=list(boundary_ids))
                with open(STATE_FILE, "w") as f:
                    json.dump(state, f)
                gc.collect()
    finally:
```

```python
        # flush whatever's left even if we broke out early, never lose a partial batch
        if batch_rows:
            pd.DataFrame(batch_rows).to_csv(ALL_SWATHS_CSV, mode="a", header=write_header, index=False)
        if acquisitions_batch:
            pd.DataFrame(acquisitions_batch).to_csv(ALL_ACQUISITIONS_CSV, mode="a", header=write_header_acq, index=False)
        state.update(n_scanned=n_new, seen_grid_codes=list(seen_grid_codes),
                      last_datetime=last_seen_datetime, boundary_ids=list(boundary_ids))
        with open(STATE_FILE, "w") as f:
            json.dump(state, f)

    if page_error and not finished:
        time.sleep(OUTER_RECONNECT_DELAY)  # let the API recover before hammering it again

print(f"{len(seen_grid_codes)} unique swaths from {n_new} granules scanned.")
```

Listing 4: The scan loop, safe to interrupt and rerun as it starts from point of last use.

C.3.4 Diagnostics: is the catalog actually representative?

Discovery here is stepped, not smoothly saturating: long flat plateaus of repeat visits punctuated by unpredictable bursts of new swaths. Roughly 11% of the final unique-swath total in this run turned up in just the last 15% of the scan, so a partial scan can't be assumed representative just because the curve looks flat for a while.

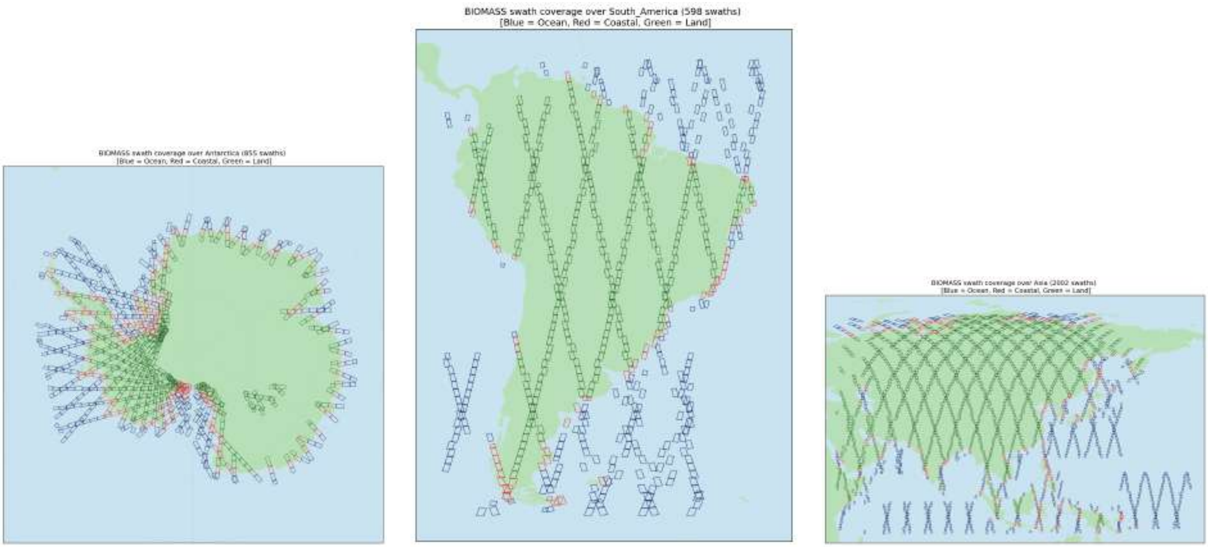
```python
import matplotlib.pyplot as plt
from datetime import datetime

acq_df = pd.read_csv(ALL_ACQUISITIONS_CSV)
acq_df["datetime"] = pd.to_datetime(acq_df["datetime"])
acq_df = acq_df.sort_values("datetime").reset_index(drop=True)  # must be chronological for "cumulative" to mean anything
acq_df["is_first_occurrence"] = ~acq_df["grid_code"].duplicated()
acq_df["cumulative_unique"] = acq_df["is_first_occurrence"].cumsum()

fig, ax = plt.subplots(figsize=(10, 6))
ax.plot(range(1, len(acq_df) + 1), acq_df["cumulative_unique"], linewidth=1.2)
ax.set_xlabel("Items scanned")
ax.set_ylabel("Unique swaths discovered")
```

```
ax.set_title(f"Biomass L1A SLC new paths discovered as of {datetime.now
    ():%d-%m-%y}")
plt.tight_layout()
plt.show()
```

Listing 5: Discovery curve: cumulative unique swaths vs cumulative items scanned.

C.3.5 Classified world map

This is the most visual element providing information on the full swath locations and should always be included for clarity. This section reads `all_swaths.csv` fresh off disk, classifies each unique swath by `ocean_fraction` against the stated thresholds, and plots the result. Coastlines are Natural Earth 50m rather than the coarser 110m default, mainly for Antarctica's sake, as altering this avoided truncated and entirely misleading visuals. Also, it is helpful to visualize the product paths in their entirety, as ESA states Biomass's P-band radar is switched off within the line-of-sight of US Space Object Tracking Radar stations, "mainly excluding North America and Europe." However, this is only published as a graphic, not coordinates; as such, this cell provides a cleaner visual. This code was also extended to display only coverage within a given continent or country of interest as shown in the figures below.



(a) Antarctica (b) South America (c) Asia

Figure 14: Regional Biomass swath coverage over Antarctica, South America, and Asia, generated by the region/country-zoom cell of the swath scanner (Appendix C.3), with swaths coloured by ocean/coastal/land classification.

```
from shapely import wkt as shapely_wkt
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import cartopy.feature as cfeature
```

```python

swaths_geo = pd.read_csv(ALL_SWATHS_CSV).drop_duplicates(subset="grid_code").copy()
swaths_geo["geometry"] = swaths_geo["geometry_wkt"].apply(shapely_wkt.loads)
ocean_df = swaths_geo[swaths_geo["ocean_fraction"] >= OCEAN_THRESHOLD]
coastal_df = swaths_geo[(swaths_geo["ocean_fraction"] > LAND_ONLY_THRESHOLD) & (swaths_geo["ocean_fraction"] < OCEAN_THRESHOLD)]
land_df = swaths_geo[swaths_geo["ocean_fraction"] <= LAND_ONLY_THRESHOLD]

def plot_ring_dateline_safe(ax, xs, ys, **kwargs):
    """Splits a ring wherever it jumps across the antimeridian, so it doesn't
    draw a spurious line straight across the map."""
    xs, ys = list(xs), list(ys)
    jumps = [i for i in range(1, len(xs)) if abs(xs[i] - xs[i - 1]) > 180]
    start = 0
    for j in jumps + [len(xs)]:
        if j - start >= 2:
            ax.plot(xs[start:j], ys[start:j], transform=ccrs.PlateCarree(), **kwargs)
        start = j

fig = plt.figure(figsize=(12, 20))
ax = plt.axes(projection=ccrs.PlateCarree())
ax.set_extent([-180, 180, -81, 90], crs=ccrs.PlateCarree())  # crops the far south, otherwise Antarctica's dense tangle dominates
ax.add_feature(cfeature.NaturalEarthFeature('physical', 'ocean', '50m', facecolor="#c9e3f0", edgecolor="none"), zorder=0)
ax.add_feature(cfeature.NaturalEarthFeature('physical', 'land', '50m', facecolor="#b8e0b8", edgecolor="none"), zorder=0)

# draw order: land, then ocean, then coastal on top, so coastal swaths don't get buried
for df, color in [(land_df, "#1a5c1a"), (ocean_df, "#0a1a5c"), (coastal_df, "#d62728")]:
    for geom in df["geometry"]:
        geoms = [geom] if geom.geom_type == "Polygon" else list(geom.geoms)
        for g in geoms:
            xs, ys = g.exterior.xy
            plot_ring_dateline_safe(ax, xs, ys, color=color, linewidth=0.7, alpha=0.8)

```

```
ax.set_title(f"Ocean: {len(ocean_df)} | Land: {len(land_df)} | Coastal: {len(coastal_df)}")
plt.tight_layout()
plt.savefig(OUTPUT_DIR / "ocean_land_coastal_swaths.png", dpi=150, bbox_inches="tight")
plt.show()
```

Listing 6: World map, all swaths classified ocean/coastal/land.

.

C.4 BIOMASS L1a Product Downloader

Individual BIOMASS granules can be downloaded directly through the ESA MAAP Explorer web interface, which offers each product as a single zip archive containing every associated file (amplitude and phase GeoTIFFs, annotation and orbit XML, a KML footprint, and a handful of others). This was the approach used earlier in the internship for convenience, but it doesn't scale well once more than a handful of granules are needed at once, particularly since only a subset of those files, generally the amplitude/phase pair and their metadata, are actually required for processing, with the rest just adding to the download size and time. The notebook cells below replace that manual step.

C.4.1 Geolocation based Product Download Cell

This cell finds candidate granules by geographic location using a bounding box rather than requiring the exact product ID from MAAP Explorer in advance.

```
from pathlib import Path
import requests
from pystac_client import Client

# either set a centre point + a buffer around it, or set BBOX directly
# (min_lon, min_lat, max_lon, max_lat), whichever's easier for the area
CENTRE_LAT, CENTRE_LON = 6.8, -58.2 # e.g. Demerara River Delta
BUFFER_DEG = 0.5 # ~55km at the equator, widen/narrow as needed
BBOX = None # set this to override the point+buffer above

DATE_RANGE = None # e.g. "2024-01-01/2024-12-31", or None for no date filter
MAX_RESULTS = 20

TOKEN_URL = 'https://iam.maap.eo.esa.int/realms/esa-maap/protocol/openid-connect/token'
CATALOG_URL = 'https://catalog.maap.eo.esa.int/catalogue/'

SECRETS_PATH = Path(r'C:\Users\Orlaith.Doyle\STEP_BACK') / 'eo_credentials.txt'
```

```python
exec(SECRETS_PATH.read_text())

resp = requests.post(TOKEN_URL, data={
    'client_id': maap_client_id,
    'client_secret': maap_client_secret,
    'grant_type': 'refresh_token',
    'refresh_token': maap_offline_token,
    'scope': 'offline_access openid',
})
resp.raise_for_status()
access_token = resp.json()['access_token']
print('Access token retrieved successfully.\n')

catalog = Client.open(CATALOG_URL, headers={'Authorization': f'Bearer {access_token}'})

search_bbox = BBOX or (CENTRE_LON - BUFFER_DEG, CENTRE_LAT - BUFFER_DEG,
                        CENTRE_LON + BUFFER_DEG, CENTRE_LAT + BUFFER_DEG)
search = catalog.search(
    collections=['BiomassLevel1a'],
    bbox=search_bbox,
    datetime=DATE_RANGE,
    max_items=MAX_RESULTS,
)
results = list(search.items())

print(f'{len(results)} granule(s) found intersecting bbox {search_bbox}'
      + (f' between {DATE_RANGE}' if DATE_RANGE else ' (no date filter)'))
print()
for item in results:
    print(f"{item.id}")
    print(f"datetime: {item.properties.get('datetime')}")
    print(f"orbit_state: {item.properties.get('sat:orbit_state')}")
    print(f"product_type: {item.properties.get('product:type')}")
    print()

#  for speciifc product ID filters from MAAP edit this line
TARGET_GRANULE_IDS = [item.id for item in results]
print(f'TARGET_GRANULE_IDS set with {len(TARGET_GRANULE_IDS)} granule(s), '
      f'ready for the downloader cell below.')
```

Listing 7: Locates and downloads granules by geographic location.

C.4.2 Product ID based Data Download Cell

This cell checks using product ID and bounding box location what's already been downloaded locally before pulling only the specific assets needed, skipping anything already on disk and picking back up cleanly if a download gets interrupted partway through.

```
from pathlib import Path # file/folder path handling
import requests # HTTP calls MAAP token exchange and downloading granule assets
from tqdm import tqdm # progress bar for the (large, 200-300 MB) tiff downloads
from pystac_client import Client # STAC client for searching the MAAP BIOMASS catalog by granule ID

# scene / search config
# TARGET_GRANULE_IDS could be already set from the geo-location search, or paste ID(s) directly here, IDs are copyable straight out of the MAAP explorer if you already know exactly which granule you want
if 'TARGET_GRANULE_IDS' not in dir():
    TARGET_GRANULE_IDS = [
        'BIO_S2_SCS_1S_2026073...',
        'BIO_S2_SCS_1S_2026043...',
        # add as many as needed
    ]
TOKEN_URL = 'https://iam.maap.eo.esa.int/realms/esa-maap/protocol/openid-connect/token'
CATALOG_URL = 'https://catalog.maap.eo.esa.int/catalogue/'
WANTED_ASSETS = ['i_abs', 'i_phase', 'annot_xml', 'orb_xml', 'kml']  # confirm these match asset keys in the STAC item

# load MAAP credentials from eo_credentials.txt
SECRETS_PATH = Path(r'C:\Users\Orlaith.Doyle\STEP_BACK') / 'eo_credentials.txt'
exec(SECRETS_PATH.read_text())

# where to look for / save granules
SIDEQUEST_DIR = Path(r'C:\Users\Orlaith.Doyle\STEP_BACK') / 'sidequest_check' # search here first, older downloads live here
OUT_DIR = Path(r'C:\Users\Orlaith.Doyle\STEP_BACK') / 'DemeraraRiverDelt' # new downloads go here, obviously change it according to your ROI!
OUT_DIR.mkdir(parents=True, exist_ok=True)
def download_asset(url, dest, token):
    if dest.exists():
        print(f' Already exists: {dest.name}')
        return
    headers = {'Authorization': f'Bearer {token}'}
```

```python
    with requests.get(url, headers=headers, stream=True, timeout=600) as r:
        r.raise_for_status()
        total = int(r.headers.get('content-length', 0))
        with open(dest, 'wb') as f, tqdm(total=total, unit='B', unit_scale=True,
                                          desc=dest.name[:55], leave=False) as bar:
            for chunk in r.iter_content(chunk_size=8 << 20):
                f.write(chunk)
                bar.update(len(chunk))
    print(f'    Saved: {dest.name} ({dest.stat().st_size / 1e6:.1f} MB)')

def find_local_dir(granule_id):
    '''Checks SIDEQUEST_DIR first (older downloads), then OUT_DIR (this
    notebook's own downloads), matching either the granule ID's original
    or lowercased form.'''
    for base in (SIDEQUEST_DIR, OUT_DIR):
        for candidate in (base / granule_id, base / granule_id.lower()):
            if candidate.exists():
                return candidate
    return None

granule_paths = {}
missing = []

for gid in TARGET_GRANULE_IDS:
    local_dir = find_local_dir(gid)
    if local_dir is not None and next(local_dir.glob('*i_abs.tiff'), None) is not None:
        granule_paths[gid] = {
            'nom_dir': local_dir,
            'annot_xml': next(local_dir.glob('*annot.xml'), None),
            'orb_xml': next(local_dir.glob('*orb.xml'), None),
            'kml': next(local_dir.glob('*.kml'), None),
            'amp_tif': next(local_dir.glob('*i_abs.tiff'), None),
            'pha_tif': next(local_dir.glob('*i_phase.tiff'), None),
        }
        print(f'  Found in {local_dir.parent.name}: {gid}')
    else:
        missing.append(gid)

print(f'\nAlready have: {len(granule_paths)} of {len(TARGET_GRANULE_IDS)}')
print(f'Missing: {len(missing)} of {len(TARGET_GRANULE_IDS)}')

```

```python
if missing:
    # get access token
    resp = requests.post(TOKEN_URL, data={
        'client_id': maap_client_id,
        'client_secret': maap_client_secret,
        'grant_type': 'refresh_token',
        'refresh_token': maap_offline_token,
        'scope': 'offline_access openid',
    })
    resp.raise_for_status()
    access_token = resp.json()['access_token']
    print('\nAccess token retrieved successfully.\n')

    catalog = Client.open(CATALOG_URL, headers={'Authorization': f'Bearer {access_token}'})
    search = catalog.search(collections=['BiomassLevel1a'], ids=missing)
    items = list(search.items())
    found_ids = {item.id for item in items}
    not_found = [gid for gid in missing if gid not in found_ids]
    if not_found:
        print(f'WARNING: {len(not_found)} granule(s) not found in the catalog:')
        for gid in not_found:
            print(f'  {gid}')

    print(f'\nDownloading {len(items)} missing granule(s):\n')
    for item in items:
        print(f'=== Granule: {item.id} ===')
        nom = OUT_DIR / item.id
        nom.mkdir(parents=True, exist_ok=True)
        for key in WANTED_ASSETS:
            if key in item.assets:
                url = item.assets[key].href
                dest = nom / url.split('/')[-1]
                download_asset(url, dest, access_token)
            else:
                print(f'  NOTE: "{key}" not available for this granule.')

        granule_paths[item.id] = {
            'nom_dir': nom,
            'annot_xml': next(nom.glob('*annot.xml'), None),
            'orb_xml': next(nom.glob('*orb.xml'), None),
            'kml': next(nom.glob('*.kml'), None),
            'amp_tif': next(nom.glob('*i_abs.tiff'), None),
            'pha_tif': next(nom.glob('*i_phase.tiff'), None),
        }
```

```
        print()
else:
    print('\nNothing to download -- all target granules already found locally.')

print(f'\ngranule_paths ready with {len(granule_paths)} of {len(TARGET_GRANULE_IDS)} target granules.')
```

Listing 8: Confirms files already available on drive and then downloads anything missing from desired location/granule ID.

C.5 Baseline Product Visualisation

Before moving on to salinity retrieval, each granule was visually inspected as a first quality check. This included the multilooked intensity, the Pauli RGB image, all four polarisation channels, and the H/A/α decomposition. The aim here was to simply confirm that the ocean surface in the scene was behaving as expected under the Bragg scattering model the retrieval relies on: calm, low-entropy, surface-dominated scattering, rather than something rougher or more complex that would need separate handling before trusting the retrieval on that granule.

C.5.1 Complex SLC Reconstruct

Complex SLC Biomass L1A products store the scattering measurements as separate amplitude and phase TIFF layers, one pair per polarisation channel (HH, HV, VH, VV). Each pixel's complex scattering value is reconstructed as

$$s = A\cos(\varphi) + jA\sin(\varphi),$$

where A is the amplitude and φ is the phase. This recovers the original single-look complex (SLC) representation of the scene, which is the form required by all subsequent polarimetric processing: the 2×2 scattering matrix $\mathbf{S}$ at each pixel,

$$\mathbf{S} = \begin{pmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{pmatrix}.$$

Amplitude and phase statistics (min/max/mean) were logged for each polarisation at this stage as a basic sanity check on the raw data before any further processing.

```
    # Reconstruct complex SLC per polarisation from amplitude + phase TIFFs
slc = {}
for i, pol in enumerate(polarisations):
    slc[pol] = (matricesamp[i] * np.cos(matricespha[i])
```

```python
                + matricesamp[i] * np.sin(matricespha[i]) * 1.0j).astype(np.complex64)

# Build S-matrix xarray dataset (PolSARpro convention: y=azimuth, x=range)
ds_S = xr.Dataset(
    {pol: (["y", "x"], slc[pol]) for pol in ["hh", "hv", "vh", "vv"]},
    coords={"y": np.arange(nlines), "x": np.arange(nsamples)},
    attrs={"poltype": "S", "description": "Scattering matrix - Biomass L1A"}
)
```

Listing 9: Reconstruction of complex SLC values from amplitude/phase TIFF pairs and assembly into the 2×2 scattering matrix S.

C.5.2 Multilooking & Boxcar Averaging

The scattering matrix S is first converted to the 3×3 Hermitian coherency matrix $C3$ by forming the outer product of the Pauli-basis scattering vector k with itself,

$$C3 = \langle k\, k^{*T} \rangle, \qquad k = \Big[S_{HH},\ \sqrt{2} S_{HV},\ S_{VV} \Big]^{T}.$$

A single-look estimate of $C3$ is not statistically reliable, so multilooking is applied by averaging over 18 pixels in azimuth and 3 pixels in range, chosen to bring the effective ground resolution cell closer to square given Biomass's native pixel spacing. An additional 3×3 boxcar spatial filter is then applied on top of the multilooked result. Both operations reduce speckle noise, at the cost of spatial resolution, and are a standard prerequisite before any eigenvalue-based polarimetric decomposition (see H/A/α below), which is otherwise highly sensitive to speckle.

```python
# Multilook: looks chosen to bring azimuth/range pixel spacing to comparable ground resolution for Biomass P-band geometry
looksa, looksr = 18, 3
ds_C3    = S_to_C3(ds_S)
ds_C3_ml = multilook(ds_C3, dim_az=looksa, dim_rg=looksr)

# Additional boxcar averaging on top of multilooking, further
#     reducing speckle ahead of the H/A/alpha eigen-decomposition
ds_C3_ml_box = boxcar(ds_C3_ml, dim_az=3, dim_rg=3)
```

Listing 10: Conversion of S to the coherency matrix C_3, followed by multilooking (18×3) and boxcar filtering to reduce speckle ahead of eigen-decomposition.

C.5.3 Pauli Decomposition

The Pauli decomposition re-expresses the scattering matrix in the orthogonal Pauli basis, giving three complex coefficients that map directly onto physically distinct scattering mechanisms:

$$\begin{aligned} a &= \tfrac{1}{\sqrt{2}}(S_{HH} + S_{VV}) && \text{(surface / odd-bounce scattering)} \\ b &= \tfrac{1}{\sqrt{2}}(S_{HH} - S_{VV}) && \text{(dihedral / double-bounce scattering)} \\ c &= \sqrt{2}\, S_{HV} && \text{(volume / depolarising scattering)} \end{aligned}$$

These are mapped to red ($|b|^2$), green ($|c|^2$), and blue ($|a|^2$) to produce a false-colour Pauli RGB image. This gives an immediate qualitative read of the dominant scattering type across the scene: calm open water should appear predominantly blue (surface scattering), while vegetated land or structured targets show up red or green.

```
# Pauli RGB: R=|HH-VV| (double-bounce), G=|HV| (volume), B=|HH+VV| (
    surface)
rgb = pauli_rgb(ds_C3_ml_box, q=0.98)
rgb_display = np.transpose(rgb.values, (1, 2, 0))

fig, ax = plt.subplots(figsize=(10, 8))
ax.imshow(rgb_display, aspect='auto')
ax.set_title("Pauli RGB decomposition")
ax.set_xlabel("Range (px)"); ax.set_ylabel("Azimuth (px)")
plt.savefig("pauli_rgb.png", dpi=150, bbox_inches='tight')
\begin{lstlisting}
# Four-polarisation intensity panel (dB, 6-sigma stretch)
fig, axes = plt.subplots(1, 4, figsize=(20, 5))
for ax, pol, img in zip(axes, ["hh", "hv", "vh", "vv"],
                                [s11_png, s12_png, s21_png, s22_png]):
    ax.imshow(img, cmap='gist_gray', aspect='auto')
    ax.set_title(pol.upper())
    ax.set_xlabel("Range (px)"); ax.set_ylabel("Azimuth (px)")
plt.savefig("4pol_preview.png", dpi=150, bbox_inches='tight')
```

Listing 11: Pauli decomposition of the scattering matrix, mapped to RGB. Colour interpretation; red = double-bounce (potentially buildings or high vertical waves), green = volume scattering (Tress or fields), blue = surface scattering (Ocean, lakes, roads, runways).

C.5.4 H/A/α (Cloude-Pottier) decomposition

This is an eigenvector/eigenvalue decomposition of the multilooked $C3$ matrix at each pixel,

$$C3 = \sum_{i=1}^{3} \lambda_i \, e_i e_i^{*T}, \qquad \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0,$$

which decomposes the scattering into three orthogonal, statistically independent components. From the eigenvalues and eigenvectors, three parameters are derived:

- Entropy $H = -\sum_{i=1}^{3} p_i \log_3 p_i$, with $p_i = \lambda_i / \sum_j \lambda_j$, measuring how mixed the scattering mechanism is. $H = 0$ indicates a single, pure scattering mechanism; $H = 1$ indicates fully random, unpolarised scattering.
- Anisotropy $A = (\lambda_2 - \lambda_3)/(\lambda_2 + \lambda_3)$, which distinguishes between scenes dominated by one mechanism versus two comparably significant mechanisms, and is most informative at moderate-to-high entropy where H alone becomes ambiguous.
- Mean alpha angle $\alpha = \sum_i p_i \alpha_i$, derived from each eigenvector, indicating the scattering type on a continuum from $\alpha \approx 0°$ (surface/Bragg scattering) through $\alpha \approx 45°$ (volume scattering) to $\alpha \approx 90°$ (dihedral/double-bounce scattering).

Together, (H, A, α) give a physically interpretable, per-pixel summary of scattering behaviour independent of the sensor's absolute calibration.

```python
# H/A/alpha (Cloude-Pottier) decomposition on the multilooked, boxcar-filtered C3
ds_haa = h_a_alpha(ds_C3_ml_box, boxcar_size=[1, 1],
                   flags=("entropy", "alpha", "anisotropy"))

fig, axes = plt.subplots(1, 3, figsize=(18, 6))
im0 = axes[0].imshow(ds_haa.entropy.values,    cmap='turbo', vmin=0, vmax=1,  aspect='auto')
im1 = axes[1].imshow(ds_haa.alpha.values,      cmap='turbo', vmin=0, vmax=90, aspect='auto')
im2 = axes[2].imshow(ds_haa.anisotropy.values, cmap='jet',   vmin=0, vmax=1,  aspect='auto')
for ax, title, im in zip(axes, ["Entropy (H)", "Alpha angle (degree)", "Anisotropy (A)"],
                                [im0, im1, im2]):
    ax.set_title(title); plt.colorbar(im, ax=ax, fraction=0.046)
plt.savefig("h_a_alpha.png", dpi=150, bbox_inches='tight')
```

Listing 12: H/A/α (Cloude-Pottier) decomposition of the multilooked C_3 matrix, producing per-pixel entropy, alpha angle, and anisotropy maps.

C.5.5 H-alpha plane

The entropy and alpha values for every pixel in the scene are plotted against one another as a 2D histogram; the standard Cloude-Pottier H–α classification plane. The plane is conventionally divided into eight or nine zones corresponding to combinations of low, medium, and high entropy with surface, volume, or dihedral-dominated α, and provides a compact way of checking, at a glance, whether the scattering behaviour across a scene is consistent with expectation. For this project, open ocean pixels under the first-order Bragg assumption are expected to cluster tightly in the low-H, low-α region of the plane (zone Z9, in the standard Cloude-Pottier naming); any significant departure from that cluster is a flag that the Bragg-based forward model may not be an appropriate description for that granule, or that region of it, before proceeding to salinity retrieval.

```
# H-alpha plane: standard scatter for scattering-mechanism
    classification
ax, fig = plot_h_alpha_plane(ds_haa, bins=300, min_pts=3)
ax.set_title("H-alpha plane")
fig.savefig("h_alpha_plane.png", dpi=150, bbox_inches='tight')
```

Listing 13: H-α classification plane: 2D histogram of entropy versus alpha angle, used to verify open-ocean pixels cluster in the low-entropy, low-α Bragg-scattering zone (Z9).

C.6 HH-VV Phase Difference and SSS Chain

This code reuses the same download/authentication approach already covered and not repeated here, applied to the Amazon-mouth granules tested across two tracks. Two geometry bugs took real iteration to pin down: the ocean-mask corner assignment initially assumed a fixed `south=row0` orientation, which silently mirrored descending passes, fixed by reading `orbitPass` from the annotation XML rather than assuming a direction; and the SMAP cross-check initially returned an empty crop because SMAP's grid uses 0-360° longitude while the scene extents were in -180/+180°, fixed by detecting and converting the convention before slicing.

C.6.1 Core HH-VV Processing

```
import numpy as np, re, rasterio, pandas as pd
from datetime import datetime
from collections import defaultdict
from scipy.ndimage import binary_dilation
from global_land_mask import globe
from lxml import etree

def parse_granule_id(granule_id):
```

```python
    """Extracts start/end date, track, and frame from a BIOMASS granule ID."""
    dates = re.findall(r'(\d{8}T\d{6})', granule_id)
    start_dt = datetime.strptime(dates[0], '%Y%m%dT%H%M%S') if dates else None
    end_dt = datetime.strptime(dates[1], '%Y%m%dT%H%M%S') if len(dates) > 1 else None
    track = re.search(r'_(T\d{3})_', granule_id)
    frame = re.search(r'_(F\d{3})_', granule_id)
    return start_dt, end_dt, track.group(1) if track else None, frame.group(1) if frame else None

def get_orbit_pass(nom):
    annot_path = next(nom.glob('*annot.xml'), None)
    if annot_path is None:
        return None
    for elem in etree.parse(str(annot_path)).iter():
        if etree.QName(elem).localname == 'orbitPass' and elem.text:
            return elem.text.strip()
    return None

def _get_corner_assignment(nom):
    """Shared by build_ocean_mask and get_lonlat_grid/get_extent_from_corners
    so they always agree on row0/col0 for a given granule."""
    coords = re.search(r'<coordinates>(.*?)</coordinates>', next(nom.glob('*.kml')).read_text()).group(1)
    pts_arr = np.array([tuple(map(float, p.split(','))) for p in coords.split()])[:4, :2]

    def edge_len(a, b):
        return np.hypot(*(pts_arr[b] - pts_arr[a]))

    # shorter opposite-edge pair = range (column) direction, longer = azimuth (row). Identified by measured edge length, not compass position, so it self-corrects
    # for any swath orientation
    len01, len12, len23, len30 = edge_len(0,1), edge_len(1,2), edge_len(2,3), edge_len(3,0)
    if (len01 + len23) < (len12 + len30):
        row_groups, col_groups = ({0, 1}, {2, 3}), ({1, 2}, {0, 3})
    else:
        row_groups, col_groups = ({1, 2}, {3, 0}), ({0, 1}, {2, 3})

    row_lat_means = [(g, np.mean([pts_arr[i][1] for i in g])) for g in row_groups]
    south_group = min(row_lat_means, key=lambda x: x[1])[0]
```

```python
    north_group = max(row_lat_means, key=lambda x: x[1])[0]

    # row0 depends on orbit direction, this was confirmed empirically: Ascending = row0 = south, Descending = row0 = north, falls back to south = row0 if orbitPass is unavailable
    orbit_pass = get_orbit_pass(nom)
    row0_idx, rowmax_idx = (north_group, south_group) if orbit_pass == 'Descending' else (south_group, north_group)

    col_lon_means = [(g, np.mean([pts_arr[i][0] for i in g])) for g in col_groups]
    col0_idx = max(col_lon_means, key=lambda x: x[1])[0]
    colmax_idx = min(col_lon_means, key=lambda x: x[1])[0]

    return (pts_arr[list(row0_idx & col0_idx)[0]], pts_arr[list(row0_idx & colmax_idx)[0]],
            pts_arr[list(rowmax_idx & col0_idx)[0]], pts_arr[list(rowmax_idx & colmax_idx)[0]], orbit_pass)

def build_ocean_mask(nom, nlines, nsamples):
    row0_col0, row0_colmax, rowmax_col0, rowmax_colmax, _ = _get_corner_assignment(nom)
    row_frac, col_frac = np.linspace(0, 1, nlines)[:, None], np.linspace(0, 1, nsamples)[None, :]
    row0 = row0_col0 * (1-col_frac)[...,None] + row0_colmax * col_frac[...,None]
    rowmax = rowmax_col0 * (1-col_frac)[...,None] + rowmax_colmax * col_frac[...,None]
    grid = row0 * (1-row_frac)[...,None] + rowmax * row_frac[...,None]
    ocean_mask = globe.is_ocean(grid[...,1], grid[...,0])
    return ~binary_dilation(~ocean_mask, iterations=180)  # dilate the land side so a coastline isn't treated as clean ocean

def get_lonlat_grid(nom, nlines, nsamples):
    row0_col0, row0_colmax, rowmax_col0, rowmax_colmax, _ = _get_corner_assignment(nom)
    row_frac, col_frac = np.linspace(0, 1, nlines)[:, None], np.linspace(0, 1, nsamples)[None, :]
    row0 = row0_col0 * (1-col_frac)[...,None] + row0_colmax * col_frac[...,None]
    rowmax = rowmax_col0 * (1-col_frac)[...,None] + rowmax_colmax * col_frac[...,None]
    grid = row0 * (1-row_frac)[...,None] + rowmax * row_frac[...,None]
    return grid[..., 0], grid[..., 1]

def process_granule(granule_id, nom):
    """Load -> ocean-mask -> HH-VV phase -> circular stats."""
```

```
    amp_path, pha_path = next(nom.glob('*i_abs.tiff'), None), next(nom.glob('*i_phase.tiff'), None)
    kml_path = next(nom.glob('*.kml'), None)
    if amp_path is None or pha_path is None or kml_path is None:
        print(f'  SKIPPED {granule_id}: missing amplitude/phase tiff or KML.')
        return None

    band_order = ('VV', 'VH', 'HV', 'HH')  # BIOMASS TIFF band order is opposite of the XML label order
    with rasterio.open(amp_path) as src:
        amp = {band_order[b]: src.read(b + 1).astype(np.float32) for b in range(4)}
    with rasterio.open(pha_path) as src:
        pha = {band_order[b]: src.read(b + 1).astype(np.float32) for b in range(4)}
    SLC_HH = amp['HH'] * np.exp(1j * pha['HH'])
    SLC_VV = amp['VV'] * np.exp(1j * pha['VV'])
    nlines, nsamples = SLC_HH.shape

    ocean_mask_clean = build_ocean_mask(nom, nlines, nsamples)
    orbit_pass = get_orbit_pass(nom)

    # circular-wrapped phase difference, can't just subtract angles directly, has to go via the complex exponential or it breaks at the +pi or -pi wraparound
    phi_hh = np.where(ocean_mask_clean, np.angle(SLC_HH), np.nan)
    phi_vv = np.where(ocean_mask_clean, np.angle(SLC_VV), np.nan)
    phi_diff = np.where(ocean_mask_clean, np.angle(np.exp(1j * (phi_hh - phi_vv))), np.nan)

    valid = ~np.isnan(phi_diff)
    if valid.sum() == 0:
        print(f'  Skipped {granule_id}: no valid ocean pixels after masking.')
        return None

    # circular mean spread rather than a plain mean & std, correct for angular data
    phase_coherence = float(np.abs(np.mean(np.exp(1j * phi_diff[valid]))))
    phase_spread = float(np.sqrt(max(0.0, -2.0 * np.log(max(phase_coherence, 1e-12)))))
    circular_mean = float(np.angle(np.mean(np.exp(1j * phi_diff[valid]))))
    diff_vmin, diff_vmax = circular_mean - phase_spread, circular_mean + phase_spread
```

```python

    return {
        'phi_diff': phi_diff, 'diff_vmin': diff_vmin, 'diff_vmax': diff_vmax,
        'phase_coherence': phase_coherence, 'phase_spread': phase_spread,
        'circular_mean': circular_mean, 'nlines': nlines, 'nsamples': nsamples,
        'ocean_pct': ocean_mask_clean.mean() * 100, 'orbit_pass': orbit_pass,
    }

results = {}
for gid, paths in granule_paths.items():
    result = process_granule(gid, paths['nom_dir'])
    if result is not None:
        results[gid] = result
print(f'Processed {len(results)} of {len(granule_paths)} granules successfully.')

# summary table across every processed product
table_rows = []
for gid, res in results.items():
    start_dt, _, track, frame = parse_granule_id(gid)
    table_rows.append({
        'date': start_dt.date() if start_dt else None, 'track': track, 'frame': frame,
        'orbit_pass': res['orbit_pass'], 'ocean_pct': round(res['ocean_pct'], 1),
        'phase_coherence': round(res['phase_coherence'], 3),
        'circular_mean_rad': round(res['circular_mean'], 4),
    })
summary_df = pd.DataFrame(table_rows).sort_values(['date', 'track', 'frame']).reset_index(drop=True)
```

Listing 14: Geometry, ocean mask, and HH-VV phase difference per granule.

C.6.2 Phase Map and Histogram (min/max range)

`diff_vmin`/`diff_vmax` above are set to $\pm 1\sigma$ of the circular phase distribution. The histogram below is what that range actually looks like against the full spread, with $\pm 2\sigma$ bands shown for context.

```python
import matplotlib.pyplot as plt

def plot_phase_and_histogram(granule_id, res):
    phi_diff = res['phi_diff']
```

```python
    circular_mean, diff_vmin, diff_vmax = res['circular_mean'], res['diff_vmin'], res['diff_vmax']
    minus_2sigma, plus_2sigma = circular_mean - 2*res['phase_spread'], circular_mean + 2*res['phase_spread']
    valid = ~np.isnan(phi_diff)
    origin = 'upper' if res['orbit_pass'] == 'Descending' else 'lower'

    fig, ax = plt.subplots(figsize=(8, 10))
    im = ax.imshow(phi_diff, cmap="gist_rainbow", vmin=diff_vmin, vmax=diff_vmax, aspect="auto", origin=origin)
    ax.set_title(f"HH-VV Phase Difference; {granule_id}")
    ax.axis("off")
    fig.colorbar(im, ax=ax, fraction=0.046, pad=0.04, label="Phase (rad)")
    plt.tight_layout()
    plt.show()

    lo, hi = np.nanpercentile(phi_diff, [5, 95])
    fig, ax = plt.subplots(figsize=(10, 6))
    ax.hist(phi_diff[valid].ravel(), bins=200, range=(lo, hi), color="steelblue", alpha=0.7, zorder=3)

    band_defs = [
        (minus_2sigma, diff_vmin, "#a8d5ba", "-2sigma to -1sigma"),
        (diff_vmin, diff_vmax, "#6aa5c8", "-1sigma  to +1sigma"),
        (diff_vmax, plus_2sigma, "#a8d5ba", "+1sigma  to +2sigma"),
    ]
    for start, end, color, label in band_defs:
        ax.axvspan(max(lo, start), min(hi, end), color=color, alpha=0.3, zorder=1, label=label)
    ax.axvline(circular_mean, color="crimson", linewidth=1.5, zorder=4, label=f"circular mean = {circular_mean:.3f}")

    handles, labels = ax.get_legend_handles_labels()
    by_label = dict(zip(labels, handles))
    ax.legend(by_label.values(), by_label.keys(), loc="upper right", fontsize=8)
    ax.set_xlabel("Phase difference (rad)")
    ax.set_ylabel("Pixel count")
    ax.set_title(f"HH-VV phase difference, ocean only; {granule_id}")
    ax.grid(True, alpha=0.3, zorder=0)
    plt.tight_layout()
    plt.show()

for gid, res in results.items():
    plot_phase_and_histogram(gid, res)
```

Listing 15: Phase difference map and histogram showing the derived vmin/vmax range.

C.6.3 Mosaic into Per-Day PNG Strips

Individual BIOMASS granules only cover a single swath segment, so viewing them in isolation makes it difficult to judge whether the ocean-masked HH-VV phase difference is spatially consistent across a full repeat-pass acquisition. To address this, granules acquired on the same day (and therefore belonging to the same pass) are stitched together into a single per-day PNG strip. This gives a continuous view of the phase field along the full length of the acquisition, making it easier to spot boundary artefacts between adjacent granules and to visually assess whether the ocean mask and phase pattern hold up consistently across the whole strip, before the quantitative validation against satellite SSS in the following subsection.

```python
import cartopy.crs as ccrs
import cartopy.feature as cfeature
import io

def render_phase_plot_to_array(phase_array, cmap, vmin, vmax, orbit_pass, display_step=4):
    """Renders phi_diff to a plain RGBA image for placement on a map with
    imshow(extent=...). Downsamples first (display_step) since the full-res
    array was hitting MemoryError across a whole day's frames at once."""
    origin = 'upper' if orbit_pass == 'Descending' else 'lower'
    fig, ax = plt.subplots(figsize=(6, 8))
    ax.imshow(phase_array[::display_step, ::display_step], cmap=cmap, vmin=vmin, vmax=vmax, aspect="auto", origin=origin)
    ax.axis("off")
    fig.subplots_adjust(left=0, right=1, top=1, bottom=0)
    buf = io.BytesIO()
    fig.savefig(buf, format="png", dpi=150, transparent=True)
    plt.close(fig)
    buf.seek(0)
    return plt.imread(buf)[:, ::-1, :]  # left-right flip, confirmed needed against the quicklook

def get_extent_from_corners(nom):
    """Map extent straight from the 4 KML corners, avoids building a
    full-resolution lon/lat grid just to take its min/max."""
    corners = np.array(_get_corner_assignment(nom)[:4])
    return [corners[:, 0].min(), corners[:, 0].max(), corners[:, 1].min(), corners[:, 1].max()]

date_groups = defaultdict(list)
for gid in results.keys():
```

```
    start_dt, _, track, frame = parse_granule_id(gid)
    date_groups[start_dt.date() if start_dt else None].append({'gid': gid, 'frame': frame})

for date_key, members in sorted(date_groups.items()):
    if date_key is None:
        continue
    members_sorted = sorted(members, key=lambda m: m['frame'] or '')

    fig = plt.figure(figsize=(10, 14))
    ax = plt.axes(projection=ccrs.PlateCarree())
    all_extents = []
    for m in members_sorted:
        gid, res = m['gid'], results[m['gid']]
        extent = get_extent_from_corners(granule_paths[gid]['nom_dir'])
        img = render_phase_plot_to_array(res['phi_diff'], "gist_rainbow", res['diff_vmin'], res['diff_vmax'], res['orbit_pass'])
        all_extents.append(extent)
        ax.imshow(img, extent=extent, transform=ccrs.PlateCarree(), origin='upper', zorder=1)

    all_lons = np.concatenate([[e[0], e[1]] for e in all_extents])
    all_lats = np.concatenate([[e[2], e[3]] for e in all_extents])
    margin = 0.3
    ax.set_extent([all_lons.min()-margin, all_lons.max()+margin, all_lats.min()-margin, all_lats.max()+margin], crs=ccrs.PlateCarree())
    ax.add_feature(cfeature.NaturalEarthFeature('physical', 'land', '10m', edgecolor='black', facecolor='lightgray'), zorder=0)
    ax.set_title(f'{date_key}; {len(members_sorted)} frame(s)')
    plt.tight_layout()
    plt.show()
```

Listing 16: Same-day frames rendered and stitched onto a real map.

C.6.4 Validation against SMOS & SMAP Data

To assess whether the ocean-masked HH-VV phase difference carries a genuine oceanographic signal rather than noise, each granule's phase map is compared against an independent satellite sea surface salinity (SSS) product for the same date and location. SMOS L3 debiased SSS (CATDS CEC-LOCEAN, 9-day running mean) is used as the primary reference where available; where SMOS has no coverage for a given date or location – for example due to land/RFI contamination near the Amazon river mouth – SMAP L3 SSS (RSS, 8-day running mean) is used as a fallback. This SMOS-first approach was chosen because SMOS offers a longer observational baseline and finer native resolution over the region of interest, with SMAP providing a reliable substitute where SMOS coverage gaps occur. The two fields are plotted side by side for each granule to give a qualitative check that spatial gradients in the phase difference align with corresponding gradients in salinity.

```
# Runs the HH-VV phase vs. satellite SSS comparison across the full Amazon case (both tracks). Produces one figure per granule and a coverage summary

validation_log = []  # one row per granule: (gid, date, source_used, status)

for gid, res in results.items():
    start_dt, _, track, frame = parse_granule_id(gid)
    date_str = start_dt.strftime('%Y-%m-%d') if start_dt else None

    extent = get_extent_from_corners(granule_paths[gid]['nom_dir'])
    search_bbox = [extent[0] - 1.0, extent[2] - 1.0, extent[1] + 1.0, extent[3] + 1.0]

    sss_ds, source, sss_var = fetch_sss(date_str, search_bbox)
    if sss_ds is None:
        validation_log.append((gid, date_str, None, 'no SMOS/SMAP coverage'))
        print(f'Skipping {gid} ; no SMOS or SMAP data available near {date_str}.')
        continue

    sss_cropped = crop_to_extent(sss_ds[sss_var], extent)
    if sss_cropped.size == 0:
        validation_log.append((gid, date_str, source, 'crop empty'))
        print(f'WARNING: crop empty for {gid} ({source}).')
        continue

    # plotting
    fig, axes = plt.subplots(1, 2, figsize=(16, 8), subplot_kw={'projection': ccrs.PlateCarree()})
    img = render_phase_plot_to_array(res['phi_diff'], 'gist_rainbow', res['diff_vmin'], res['diff_vmax'], res['orbit_pass'])
    axes[0].imshow(img, extent=extent, transform=ccrs.PlateCarree(), origin='upper')
    axes[0].set_extent(extent, crs=ccrs.PlateCarree())
    axes[0].add_feature(cfeature.NaturalEarthFeature('physical', 'land', '10m', edgecolor='black', facecolor='lightgray'), zorder=0)
    axes[0].set_title(f'BIOMASS HH-VV Phase; {gid}\n{date_str} (track {track}, {frame})', fontsize=9)

    lon_for_plot = ((sss_cropped['lon'] + 180) % 360) - 180
```

```
    im1 = axes[1].pcolormesh(lon_for_plot, sss_cropped['lat'],
   sss_cropped, cmap='viridis', transform=ccrs.PlateCarree(), shading='
   auto')
    axes[1].set_extent(extent, crs=ccrs.PlateCarree())
    axes[1].add_feature(cfeature.NaturalEarthFeature('physical', 'land',
    '10m', edgecolor='black', facecolor='lightgray'), zorder=0)
    axes[1].set_title(f'{source} L3 SSS cropped to scene extent',
   fontsize=9)
    fig.colorbar(im1, ax=axes[1], fraction=0.046, pad=0.04, label='SSS (
   PSU)')
    plt.tight_layout()
    plt.savefig(f'validation_{gid}.png', dpi=150, bbox_inches='tight')
    plt.show()
    plt.close(fig)
    validation_log.append((gid, date_str, source, 'ok'))
```

Listing 17: Validation Cell against SMOS and if unavailable; SMAP datasets, to check against SSS estimate.

C.6.5 Summary plots; Mosaic, SSS Validation, Map Location Display

This final cell in the sea surface salinity experiment produces a summary figure. The figure has three panels: the mosaic HH-VV PNG strip for that date, giving a visual overview of the phase-difference signal along the swath; the corresponding SMOS or SMAP sea surface salinity field over the same extent, used as the independent validation reference (SMOS is tried first, falling back to SMAP where SMOS has no coverage for that date and location); and a small regional map with a red box marking the scene's footprint, to give geographic context at a glance. Together, these three panels allow each granule's result to be reviewed and cross-checked against satellite SSS without having to cross-reference separate outputs.

```
import matplotlib.patches as mpatches

def plot_report_panel(gid, res, granule_paths_entry, mosaic_png_path=
   None,
                          map_pad_deg=15, save=True):
    """Three-panel report figure for a single granule:
      1) the per-day HH-VV mosaic PNG strip for this date,
      2) the SMOS/SMAP SSS validation panel (SMOS-first, SMAP-fallback),
      3) a regional map with a red box marking the scene footprint.
    Reuses fetch_sss / crop_to_extent from the validation cell above.
    """
    start_dt, _, track, frame = parse_granule_id(gid)
    date_str = start_dt.strftime('%Y-%m-%d') if start_dt else None
    extent = get_extent_from_corners(granule_paths_entry['nom_dir'])
    lon_min, lon_max, lat_min, lat_max = extent
```

```python
    search_bbox = [lon_min - 1.0, lat_min - 1.0, lon_max + 1.0, lat_max + 1.0]

    if mosaic_png_path is None:
        mosaic_png_path = f'mosaics/{date_str}_hhvv_mosaic.png'  # <- swap for your actual naming

    sss_ds, source, sss_var = fetch_sss(date_str, search_bbox)

    fig = plt.figure(figsize=(18, 6))
    gs = fig.add_gridspec(1, 3)

    # Panel 1: HH-VV mosaic strip
    ax0 = fig.add_subplot(gs[0])
    try:
        ax0.imshow(plt.imread(mosaic_png_path))
        ax0.set_title(f'HH-VV Mosaic Strip\n{date_str}', fontsize=9)
    except FileNotFoundError:
        ax0.text(0.5, 0.5, 'mosaic PNG not found', ha='center', va='center', fontsize=9)
        ax0.set_title(f'HH-VV Mosaic Strip (missing)\n{date_str}', fontsize=9)
    ax0.axis('off')

    # Panel 2: SMOS/SMAP SSS validation
    ax1 = fig.add_subplot(gs[1], projection=ccrs.PlateCarree())
    ax1.add_feature(cfeature.NaturalEarthFeature('physical', 'land', '10m', edgecolor='black', facecolor='lightgray'), zorder=0)
    if sss_ds is not None:
        sss_cropped = crop_to_extent(sss_ds[sss_var], extent)
        lon_for_plot = ((sss_cropped['lon'] + 180) % 360) - 180
        im1 = ax1.pcolormesh(lon_for_plot, sss_cropped['lat'], sss_cropped, cmap='viridis', transform=ccrs.PlateCarree(), shading='auto')
        ax1.set_extent(extent, crs=ccrs.PlateCarree())
        ax1.set_title(f'{source} L3 SSS', fontsize=9)
        fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.04, label='SSS (PSU)')
    else:
        ax1.set_extent(extent, crs=ccrs.PlateCarree())
        ax1.set_title('No SMOS/SMAP coverage', fontsize=9)

    # Panel 3: location map with red box
    ax2 = fig.add_subplot(gs[2], projection=ccrs.PlateCarree())
    ax2.set_extent([lon_min - map_pad_deg, lon_max + map_pad_deg, lat_min - map_pad_deg, lat_max + map_pad_deg], crs=ccrs.PlateCarree())
```

```python
    ax2.add_feature(cfeature.NaturalEarthFeature('physical', 'land', '110m', edgecolor='black', facecolor='lightgray'), zorder=0)
    ax2.add_feature(cfeature.OCEAN, zorder=0)
    ax2.coastlines(resolution='110m')
    ax2.add_patch(mpatches.Rectangle(
        (lon_min, lat_min), lon_max - lon_min, lat_max - lat_min,
        linewidth=2, edgecolor='red', facecolor='none',
        transform=ccrs.PlateCarree(), zorder=5,
    ))
    ax2.set_title('Scene location', fontsize=9)

    fig.suptitle(f'{gid}  |  {date_str}  |  track {track}, {frame}', fontsize=11)
    plt.tight_layout()
    if save:
        plt.savefig(f'report_panel_{gid}.png', dpi=150, bbox_inches='tight')
    plt.show()
    plt.close(fig)

for gid, res in results.items():
    plot_report_panel(gid, res, granule_paths[gid])
```
Listing 18: Summary display of CPD mosaic, SSS Validation & Product location.